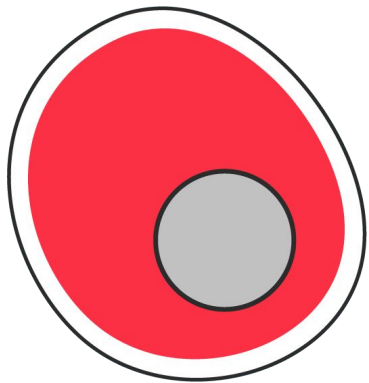




Open Problems in Single-Cell Analysis

Creating a benchmark MVP in 60 minutes 🤖

The core team



Daniel B. Burkhardt, PhD.
Co-founder
Machine Learning Scientist at
Cellarity



Robrecht Cannoodt, PhD
Core Developer
Data Science Engineer at Data
Intuitive



Scott Gigante, PhD
Co-founder
Founding Engineer at Sightline
Climate



Christopher Lance
PhD Student at Helmholtz
Zentrum München



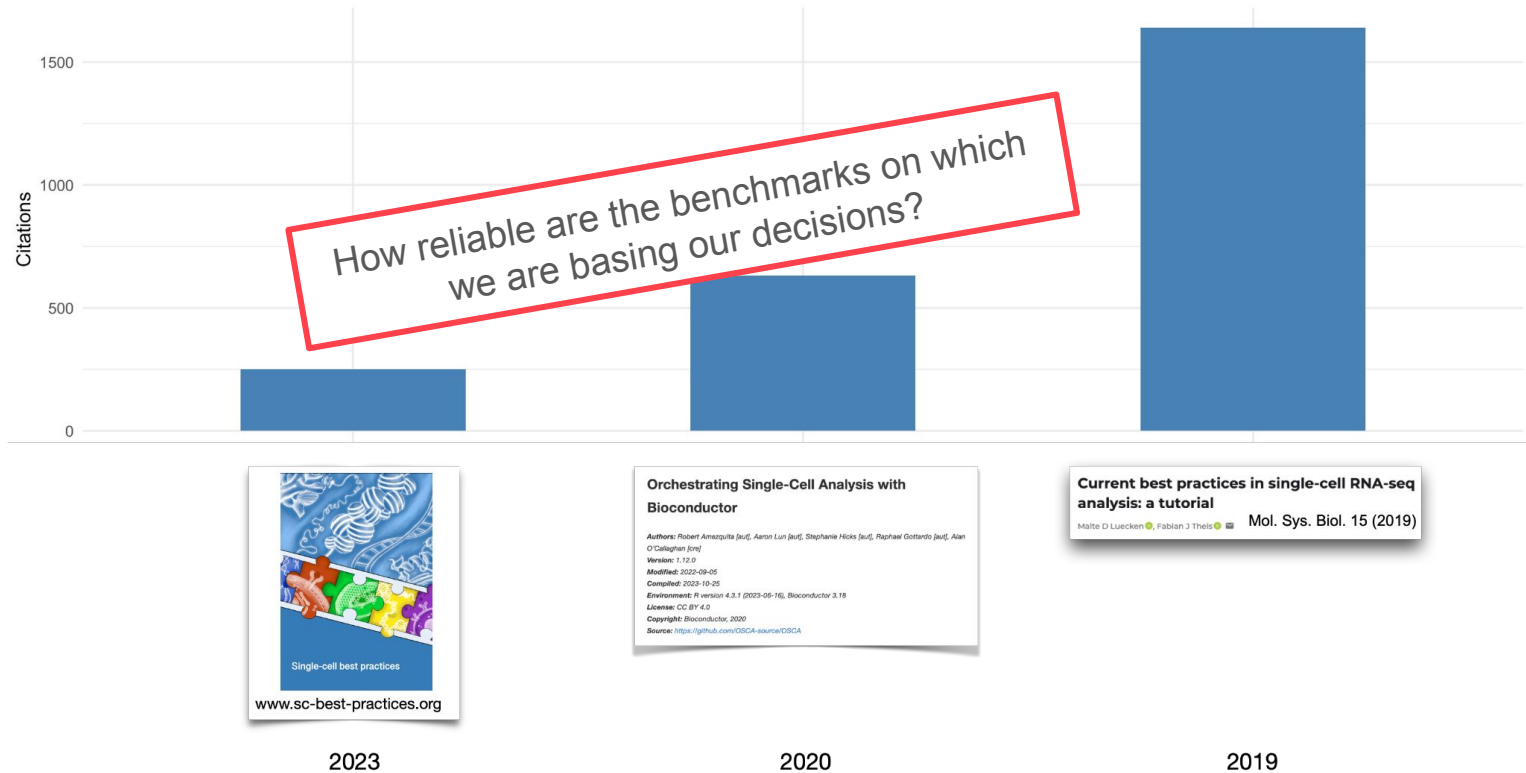
Malte D. Luecken, D.Phil
Co-founder
Group Leader at Helmholtz
Zentrum München



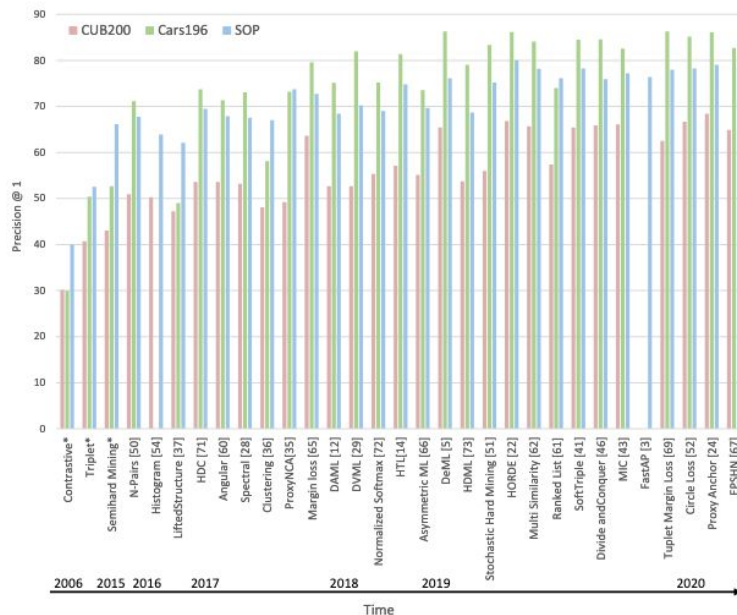
Angela O. Pisco, PhD
Director of Computational
Biology at insitro



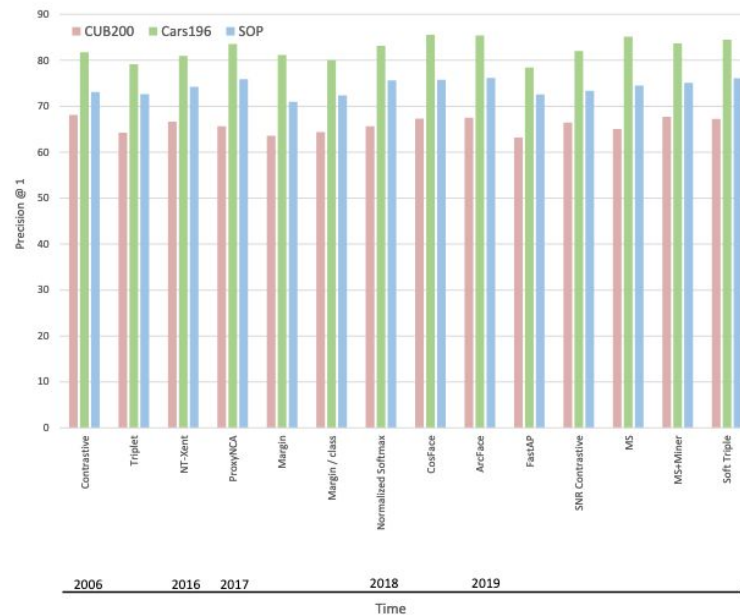
Benchmarking is driving analysis decisions in single-cell genomics



Decentralized benchmarking exaggerates improvements



(a) The trend according to papers



(b) The trend according to reality

Used metrics

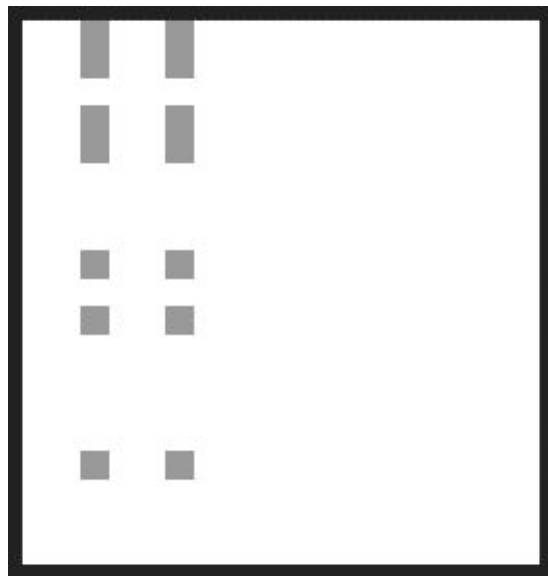
integration benchmarks

Used metrics

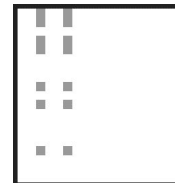
API Batch ASW
Batch entropy
Cell cycle conservation
Cell type ASW
Cell type entropy
Clustering ASW
Graph cLISI
Graph connectivity
Graph lLISI
HVG conservation
Isolated label F1
Isolated label silhouette
kBET
LISI combined
NMI
PCR batch
Trajectory conservation

Benchmarked methods

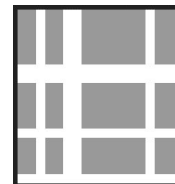
BBKNN
ComBat
DESC
fastMNN
Harmony
Liger
Limma
MDD-ResNet
mnnCorrect
Saucie
Scanorama
scanVI
scGen
scMerge
scVI
Seurat v2
Seurat v3
trVAE
ZINB-WaVE



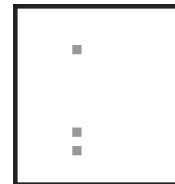
Chazarra-Gil et al., NAR (2021)



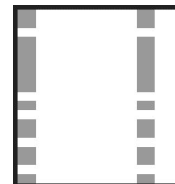
Chazarra-Gil et al., NAR (2021)



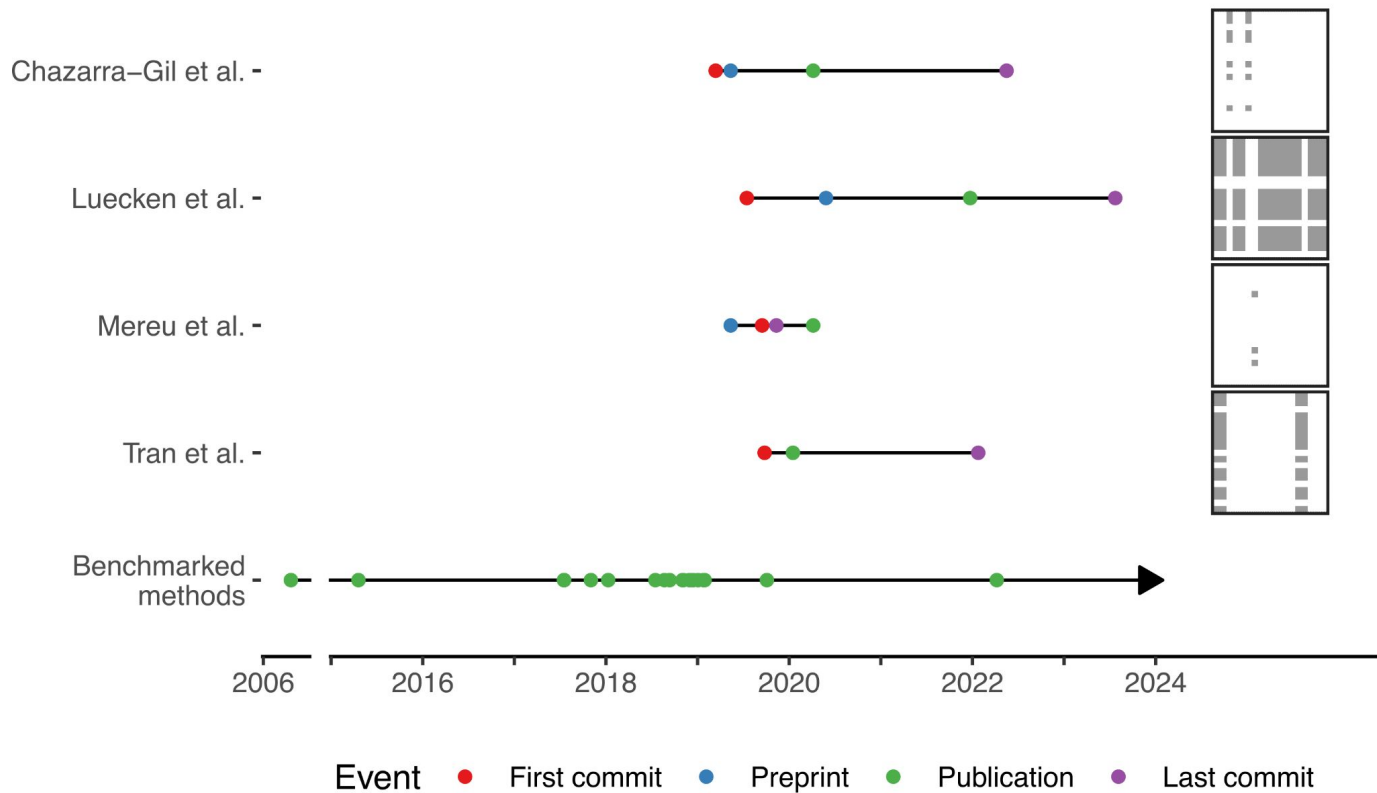
Luecken et al., Nat Meth (2022)



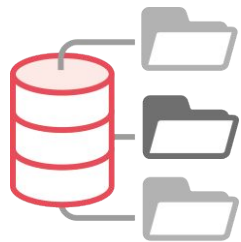
Mereu et al., Nat Biotech (2020)

Tran et al., *Genome Biol* (2021)

Benchmarks inevitably age



Addressing benchmarking limitations in Open Problems



Code
repository



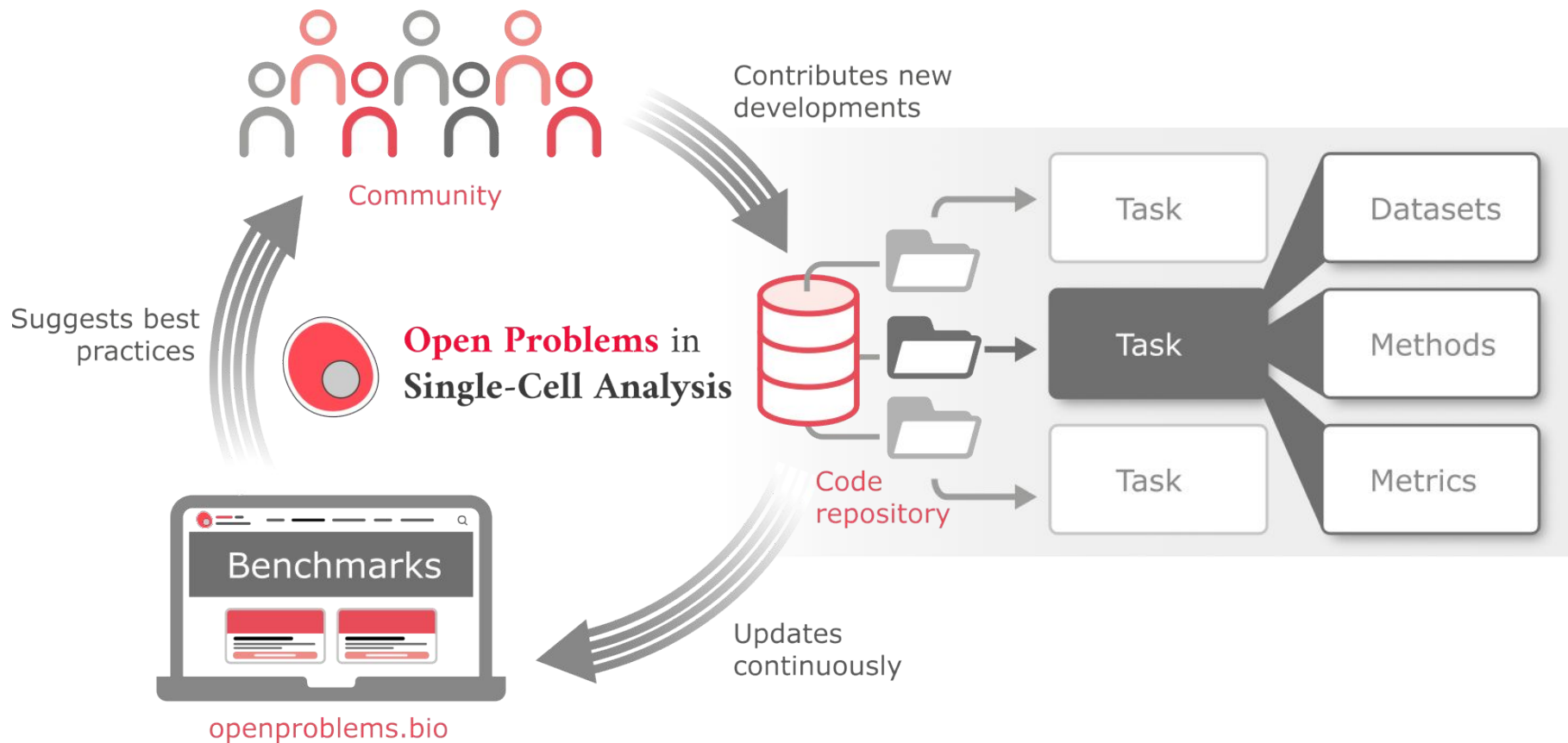
Community



openproblems.bio

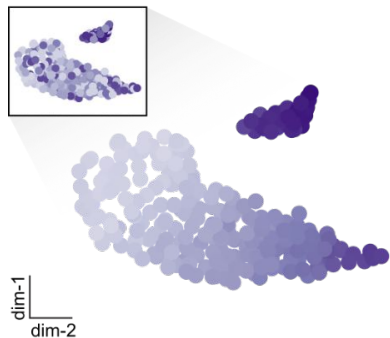
Centralization	✓		
Consistency	✓		
Longevity	✓	✓	
Accessibility			✓

The Open Problems setup

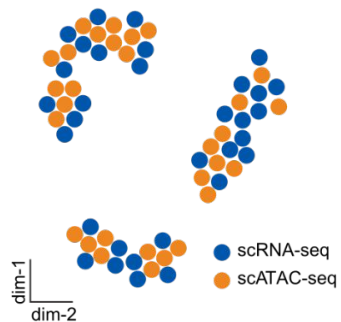


Community input defines standardized benchmarks

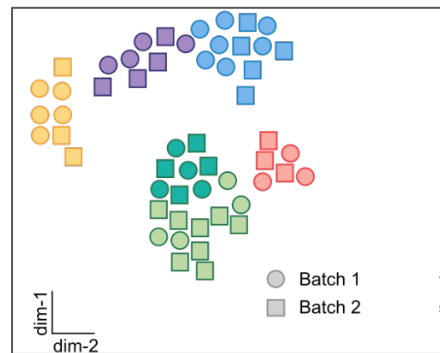
Data denoising



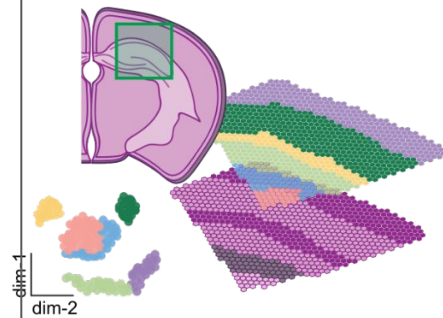
Multimodal integration



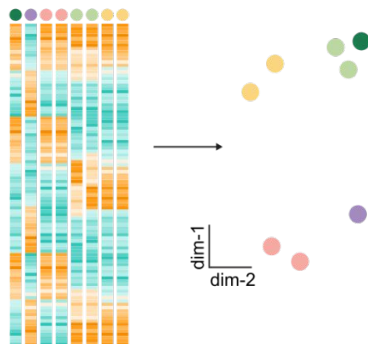
Batch integration



Spatial deconvolution



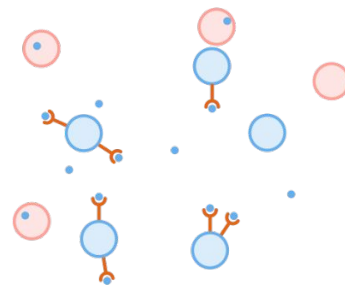
Dimensionality reduction



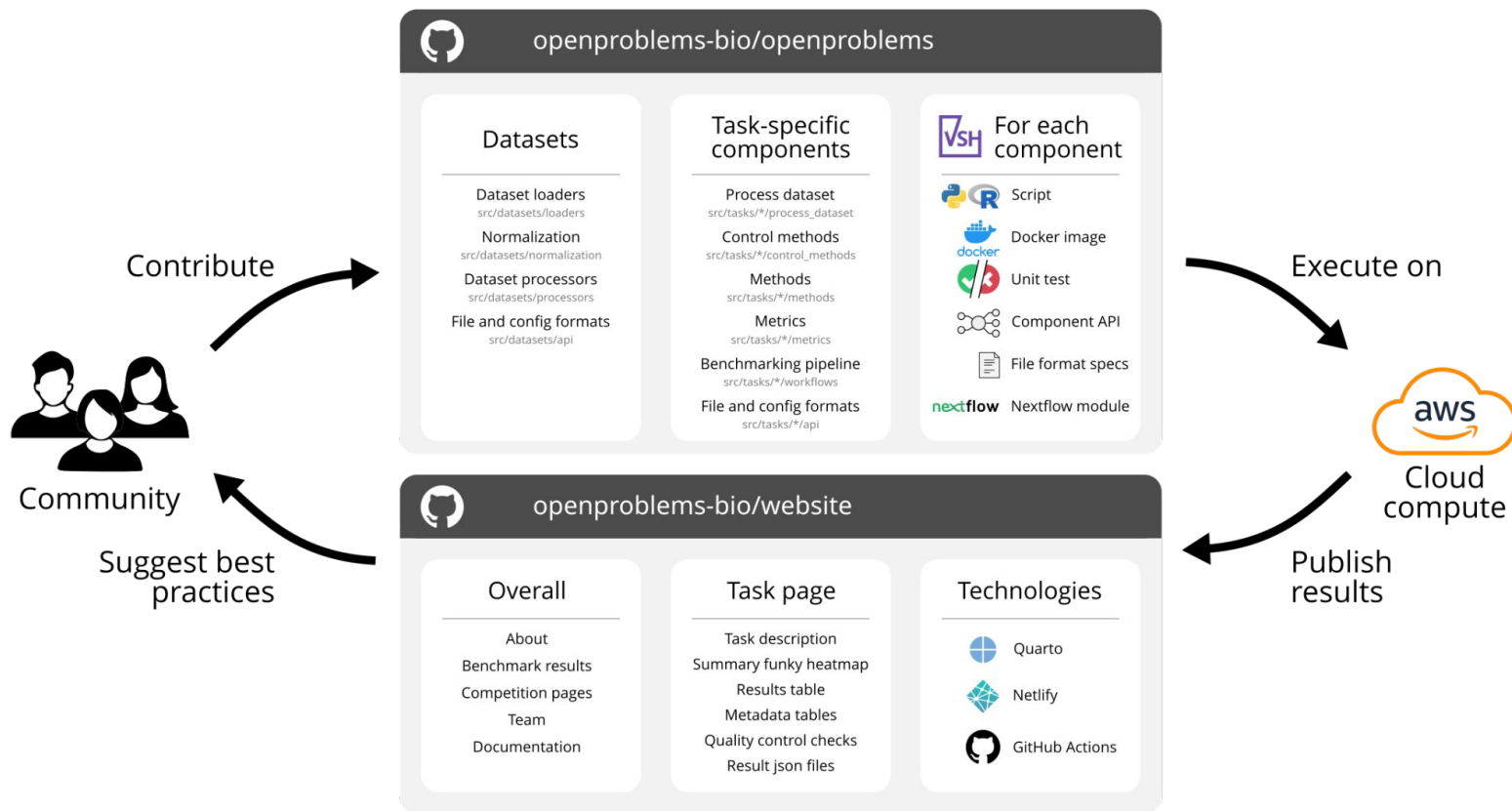
Label projection



Cell-cell communication



A look under the hood...

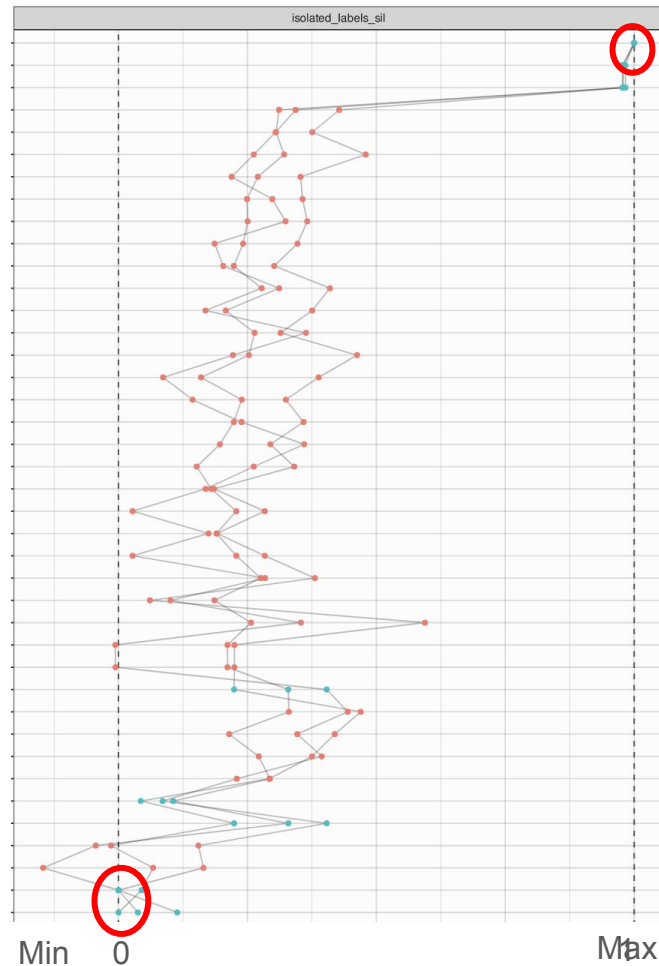


Accessibility: Ensuring comparable results across metrics via normalization

Challenge: metrics have different scales

Solution: Normalization

- Add (several) negative and positive baseline methods
 - E.g., random integration & cell type encoding as embedding
- Scale metric outputs to 0 - 1 between min and max baselines
- Investigate method outliers



<https://openproblems.bio/results/>

Batch integration embed
 Removing batch effects while preserving biological variation (embedding output)
 3 datasets • 32 methods • 8 control methods • 10 metrics

Batch integration feature
 Removing batch effects while preserving biological variation (feature output)
 3 datasets • 18 methods • 7 control methods • 11 metrics

Batch integration graph
 Removing batch effects while preserving biological variation (graph output)
 3 datasets • 40 methods • 5 control methods • 4 metrics

Cell-Cell Communication Inference (Ligand-Target)
 Detect interactions between ligands and target cell types
 1 datasets • 14 methods • 2 control methods • 2 metrics

Cell-Cell Communication Inference (Source-Target)
 Detect interactions between source and target cell types
 1 datasets • 14 methods • 2 control methods • 2 metrics

Denoising
 Removing noise in sparse single-cell RNA-seq count data
 3 datasets • 11 methods • 2 control methods • 2 metrics

		Overall	Datasets		Metrics					Resources									
Name	Score	Error reason	Immune (by batch)	Lung (Vera Briga et al.)	Pancreas (by batch)	AMR	Batch ASW	Cel Cycle Score	Graph connectivity	Isolated label F1	KD	NMI	PTC	Regression Silhouette	APPU	Peak memory	Disk read	Disk write	Duration
Random Embedding by Celltype																2G	1G	1G	7m
Random Graph by Celltype																2G	1G	1G	6m
Random Embedding by Celltype (with jitter)																2G	1G	1G	7m
scANVI (hvg/unscaled)																9G	1G	211M	2h
scANVI (full/unscaled)																6G	1G	1G	5h
MNN (hvg/scaled)																21G	3G	2G	34m
Combat (hvg/unscaled)																4G	1G	806M	11m
scVI (full/unscaled)																4G	1G	1G	5h

Timeline - First release of each task¹

Date	Repository	Version	Task
2020-07	OpenProblems	v0.1	Label Projection
2020-07	OpenProblems	v0.1	Match Modality
2021-02	OpenProblems	v0.3.1	Regulatory Effect Prediction
2021-02	OpenProblems	v0.3.1	Denoising
2021-02	OpenProblems	v0.3.1	Dimensionality Reduction
2021-07	NeurIPS 2021	v0.1	Predict Modality
2021-07	NeurIPS 2021	v0.1	Joint Embedding
2022-05	OpenProblems	v0.3.4	Batch Integration
2022-08	OpenProblems	v0.5.0	Spatial Decomposition
2022-10	OpenProblems	v0.5.7	Cell-Cell Communication
2024-06	NeurIPS 2023 Perturbation Prediction	v1.0.0	Perturbation Prediction
2024-09	OpenProblems	v2.0.0	Spatially Variable Genes
Ongoing	Spatial Tx Simulators		Spatial Tx Simulators
Ongoing	iST Preprocessing		iST Preprocessing
Ongoing	GRN Inference		GRN Inference

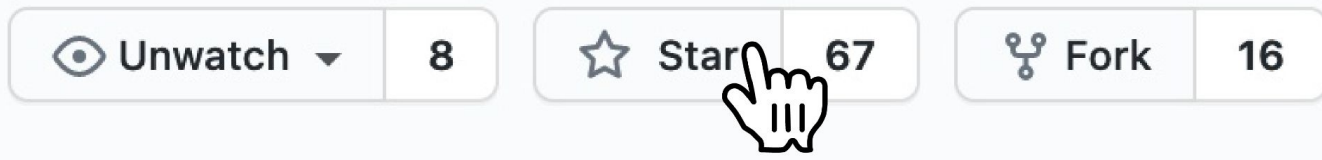
¹ This completely ignores all other improvements made to tasks and the platform that were made along the way!

A deep dive into OpenProblems

Contributing to OpenProblems

- Visit github.com/openproblems-bio/openproblems and hit the "Star" icon
- Adding datasets
- Adding methods
- Adding metrics
- Create a new benchmark from scratch
- Extending the documentation

Your first contribution? 🧑



Want to create a reusable, reproducible,
computational benchmark?

Let's get started!

... Hold on a minute! 🤔

Creating a high-quality benchmark...

- Feasibility study
- Proof of concept
- Literature study
- Minimum viable product
- Preprint
- Production benchmark
- Write manuscript
- Finishing touches
- Publication

... is just like playing Pong

Research

- Feasibility study
- Proof of concept
- Literature study
- Minimum viable product
- Preprint
- Production benchmark
- Write manuscript
- Finishing touches
- Publication

Coding

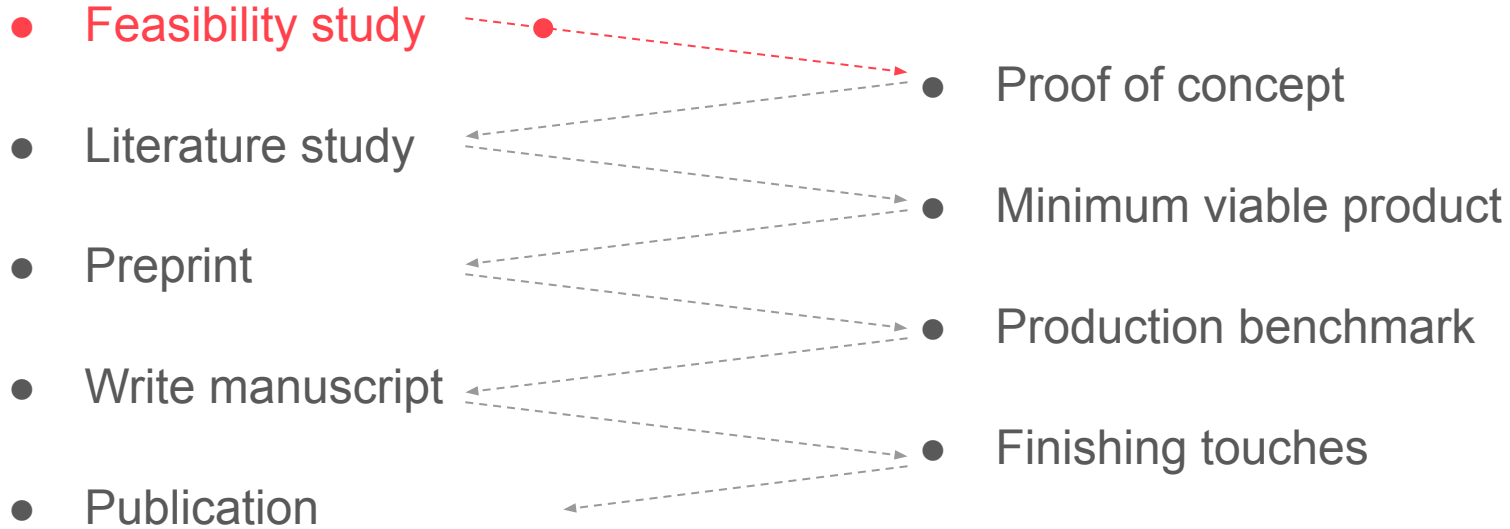
- Proof of concept
- Minimum viable product
- Production benchmark
- Finishing touches



Going back and forth is essential for gaining new insights after each step

Research

Coding



Feasibility study

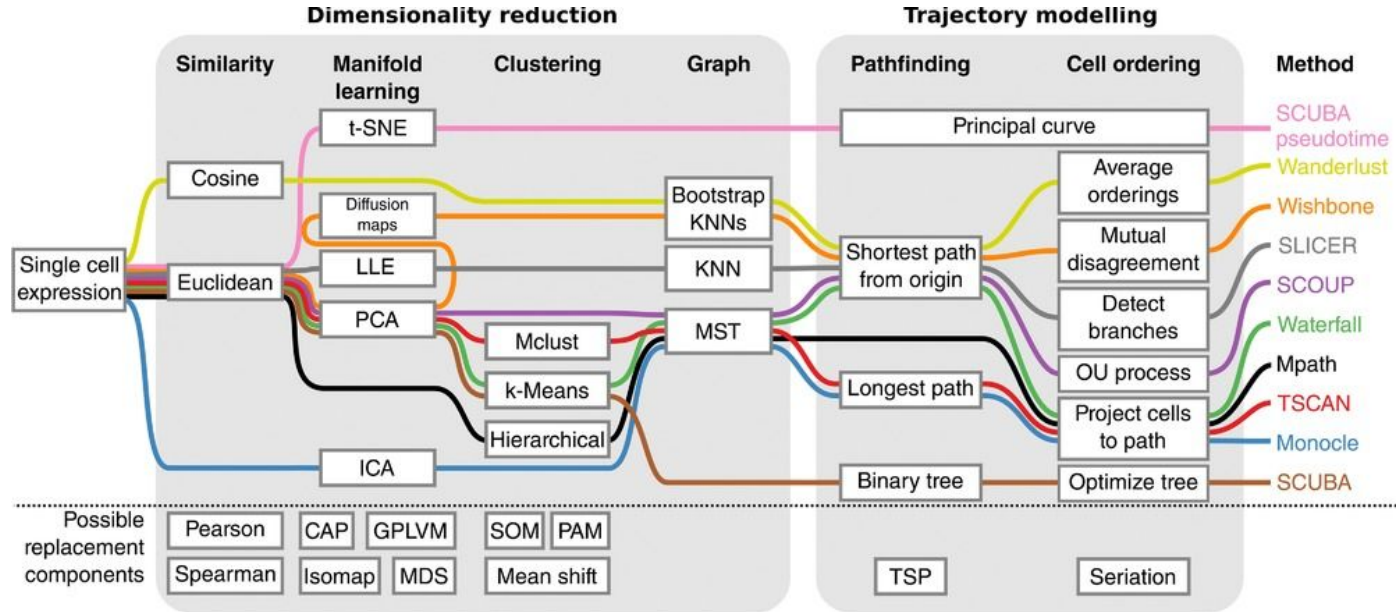
- **Formulate the problem** you're investigating
- **Find datasets** (explore the data)
- **Run a few** computational methods (follow their tutorials)
- In their manuscripts, **pay attention to:**
 - which research question they are trying to answer
 - which data structures they use
 - with which methods they compare themselves with,
 - on which datasets,
 - using which metrics

Use-case: Trajectory inference

method_id	method_components	method_directionality_infer
angle	any multiple dimensionality reductions {PCA MDS tSNE ICA LLE} + Arc-tangent	
paga	kNN + louvain clustering + connectivity test { edge statistics random walk } + diffusion pseudo time	
paga_tree	kNN + louvain clustering + connectivity test { edge statistics random walk } + diffusion pseudo time + MST	
comp1	any multiple dimensionality reductions {PCA MDS tSNE ICA LLE}	
celltree_gibbs	LDA (gibbs) + MST (on LDA topics)	
celltree_maptpx	LDA (maptpx) + MST (on LDA topics)	
celltree_vem	LDA (VEM) + MST (on LDA topics)	
dpt	diffusion map + computation of transition matrix + transition matrix to root cell + branch assignment using correlated and anti-correlated distances	root
embeddr	laplacian eigenmap + principal curves	
gpfates	GPLVM + dimensionality reduction from GPLVM + pseudotime inference from GPLVM + OMGP	
mfa	Bayesian hierarchical mixture of factor analyzers + Gibbs sampling	
monocle_ddrtree	any dimensionality reduction + (fuzzy) k-means + MST + move cells towards centers + iteration until convergence + cell projection to tree backbone	
monocle_ica	PCA + ICA + polygonal reconstruction={MST + longest shortest path (from start) + PQ tree}	root
mpath	find landmarks/waypoints + weighted neighbourhood network by number of cells + MST + MBOR ordering of cells = {hierarchical clustering}	
ouija	model expression by combining a mapping function ({switch transient}) and pseudotime, both as latent variables	
ouijaflow	model expression by combining a mapping function ({switch transient}) and pseudotime, both as latent variables	
periodpc	PCA + principal curves	
phenopath	latent variable model + mean-field co-ordinate ascent variational inference	
pseudogp	any multiple dimensionality reductions {PCA MDS tSNE ICA LLE} + GPLVM	

How it started

Literature study on trajectory inference methods



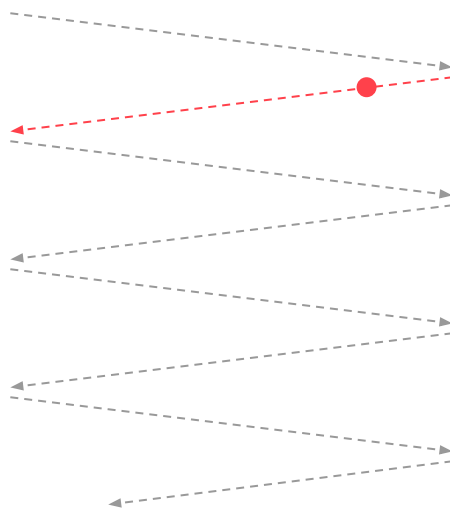
Cannoodt, R., Saelens, W. and Saeys, Y., 2016. Computational methods for trajectory inference from single-cell transcriptomics. European journal of immunology, 46(11), pp.2496-2506. doi: [10.1002/eji.201646347](https://doi.org/10.1002/eji.201646347)

Research

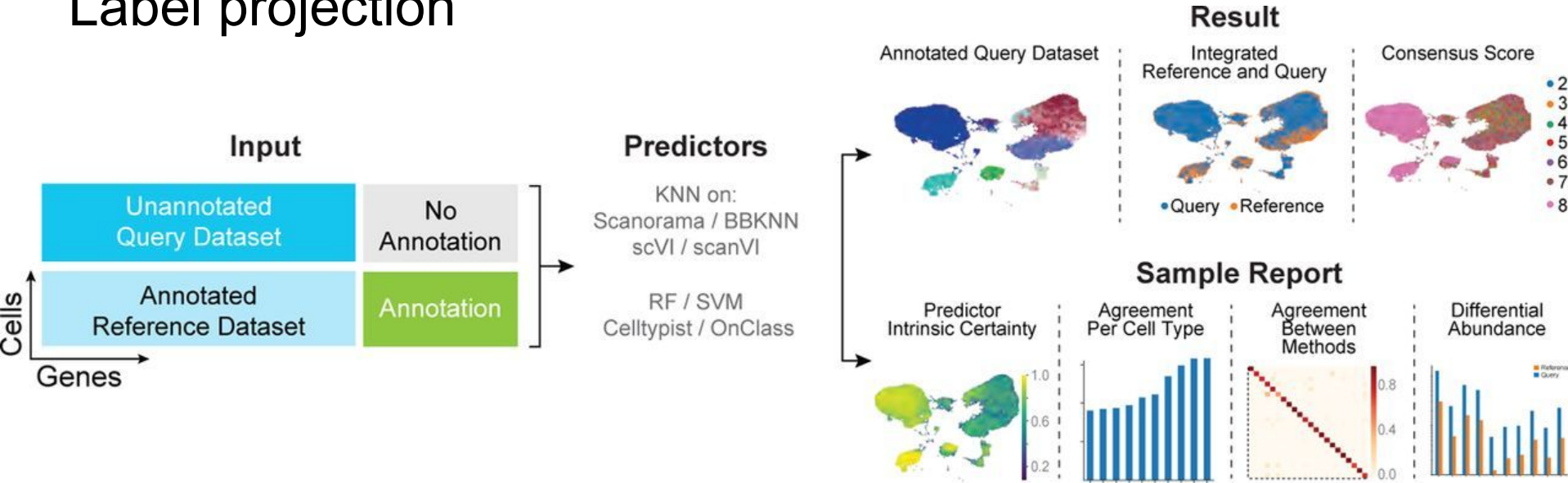
- Feasibility study
- Literature study
- Preprint
- Write manuscript
- Publication

Coding

- Proof of concept
- Minimum viable product
- Production benchmark
- Finishing touches



Label projection



Consensus prediction of cell type labels with popV

Can Ergen, Galen Xing, Chenling Xu, Michael Jayasuriya, Erin McGeever, Angela Oliveira Pisco, Aaron Streets, Nir Yosef

doi: <https://doi.org/10.1101/2023.08.18.553912>

POC script

```
import anndata as ad

# Fetch dataset from somewhere.
# -> This is downloaded from
's3://openproblems-data/resources_test/common/cxg_mouse_pancreas_atlas/dataset.h5ad'
adata = ad.read_h5ad("resources_test/common/cxg_mouse_pancreas_atlas/dataset.h5ad")

adata
# AnnData object with n_obs × n_vars = 600 × 1500
#   obs: 'dataset_id', 'assay', 'assay_ontology_term_id', 'batch', 'cell_type', ...
#   var: 'feature_id', 'feature_name', 'feature_length', 'n_cells', 'hvg', 'hvg_score'
#   uns: 'dataset_description', 'dataset_id', 'dataset_name', 'dataset_organism', 'dataset_reference',
...
#   obsm: 'X_pca'
#   layers: 'counts', 'normalized'

# Process dataset: Divide the dataset into train and test sets.
test_batch = adata.obs["batch"].to_list()[0]
train_adata = adata[adata.obs["batch"] != test_batch].copy()
test_adata = adata[adata.obs["batch"] == test_batch].copy()

# Method: Apply logistic regression
import sklearn.linear_model
classifier = sklearn.linear_model.LogisticRegression()
classifier.fit(train_adata.obsm["X_pca"], train_adata.obs["cell_type"].astype(str))
obs_label_pred = classifier.predict(test_adata.obsm["X_pca"])

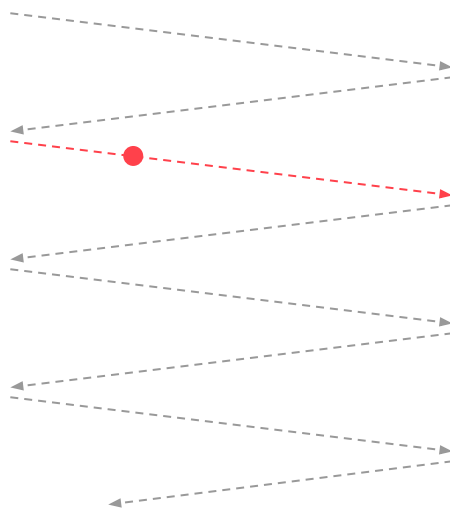
# Metric: Compute accuracy
import numpy as np
accuracy = np.mean(test_adata.obs["cell_type"] == obs_label_pred)
accuracy # 0.9478672985781991
```

Research

- Feasibility study
- Literature study
- Preprint
- Write manuscript
- Publication

Coding

- Proof of concept
- Minimum viable product
- Production benchmark
- Finishing touches



List of hypotheses & tasks

Google sheet:
[Hypotheses and tasks](#)

Hypotheses & tasks

File Edit View Insert Format Data Tools Extensions Help

Search Menus

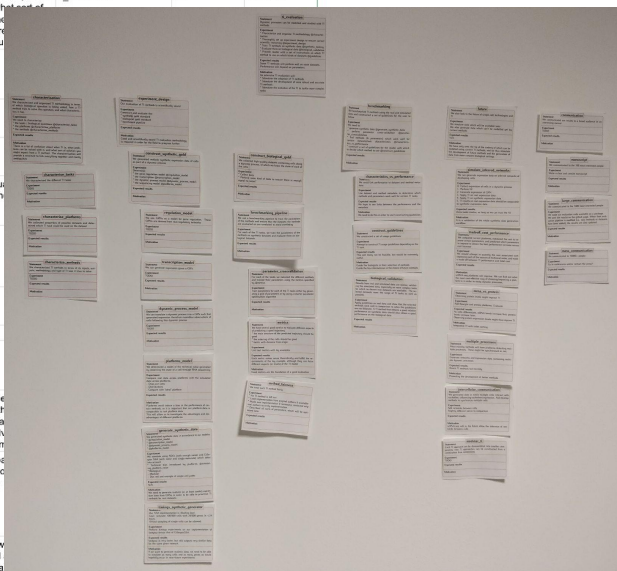
100%

10

B

I

A

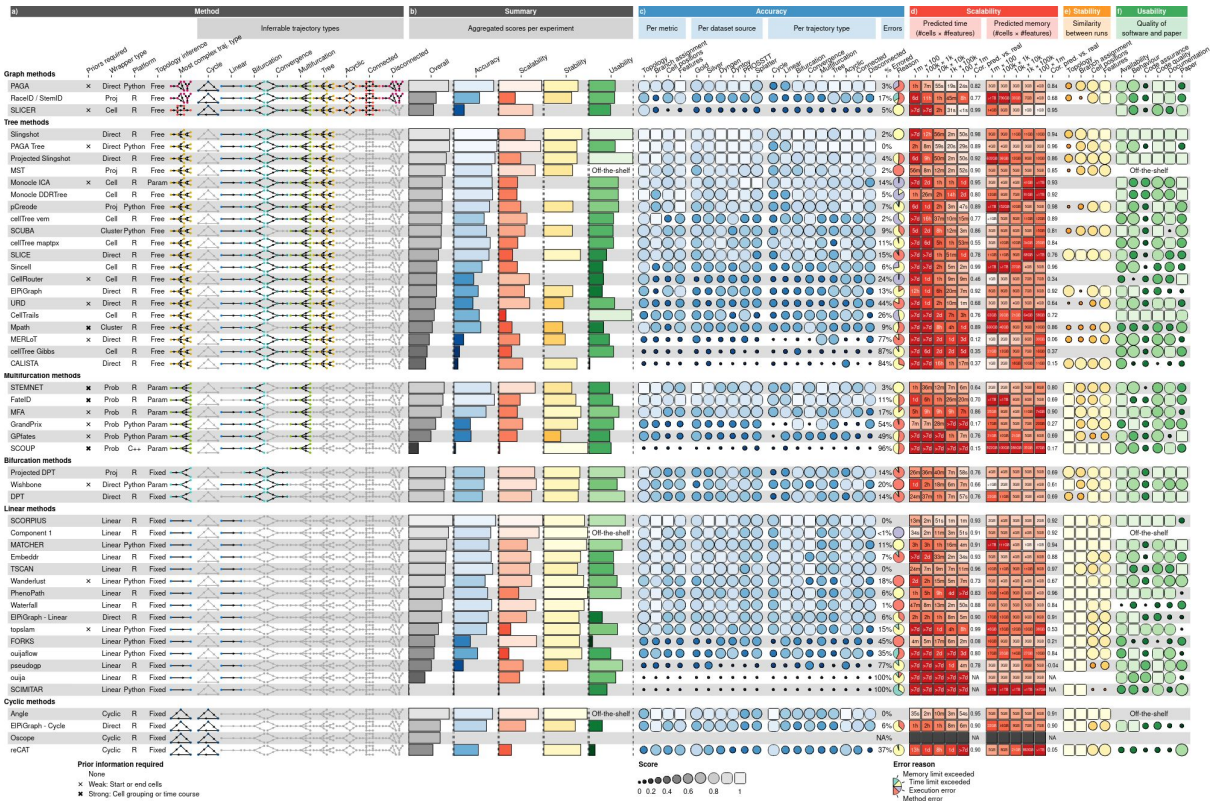


Review of manuscript and software

Google sheet:
[Overview of trajectory inference methods for dynverse](#)

[illegible]

Which resulted in a comparison of TI methods



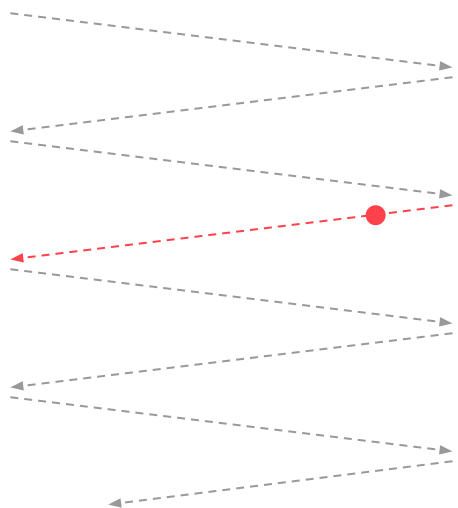
Saelens, W., Cannoodt, R., Todorov, H. and Saeys, Y., 2019. A comparison of single-cell trajectory inference methods. *Nature biotechnology*, 37(5), pp.547-554. doi: [10.1038/s41587-019-0071-9](https://doi.org/10.1038/s41587-019-0071-9)

Research

- Feasibility study
- Literature study
- Preprint
- Write manuscript
- Publication

Coding

- Proof of concept
- **Minimum viable product**
- Production benchmark
- Finishing touches



Building an MVP with OpenProblems

Building a benchmark: two approaches



A focused researcher

Has a **great idea** for a benchmark

Creates a bunch of notebooks to test hypotheses, collect results, and publish

A supplementary zip file with notebooks for running methods and metrics.

- Messy code
- No reproducibility
- Inaccurate results due to bugs

No reusability

B ioinformatics software engineer

Wants to build a **perfect benchmark**

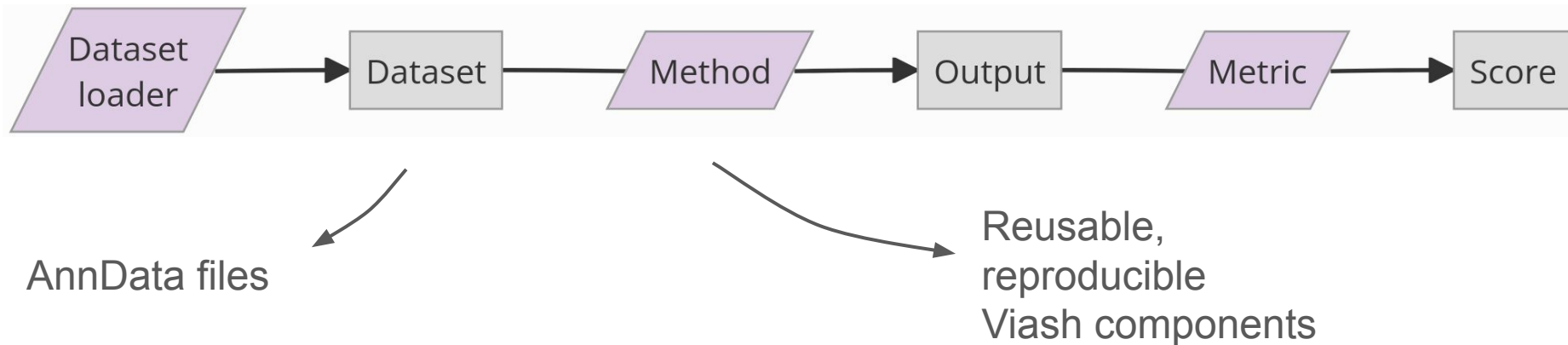
Spends an enormous amount of effort building a benchmarking framework

A new benchmarking workflow framework with dataset loaders, methods, metrics, unit tests, Docker containers, CI, CD, Linters, Docs, ...

- Massive effort
- Custom-tailored benchmark workflow
- Does not finish on time, abandons project

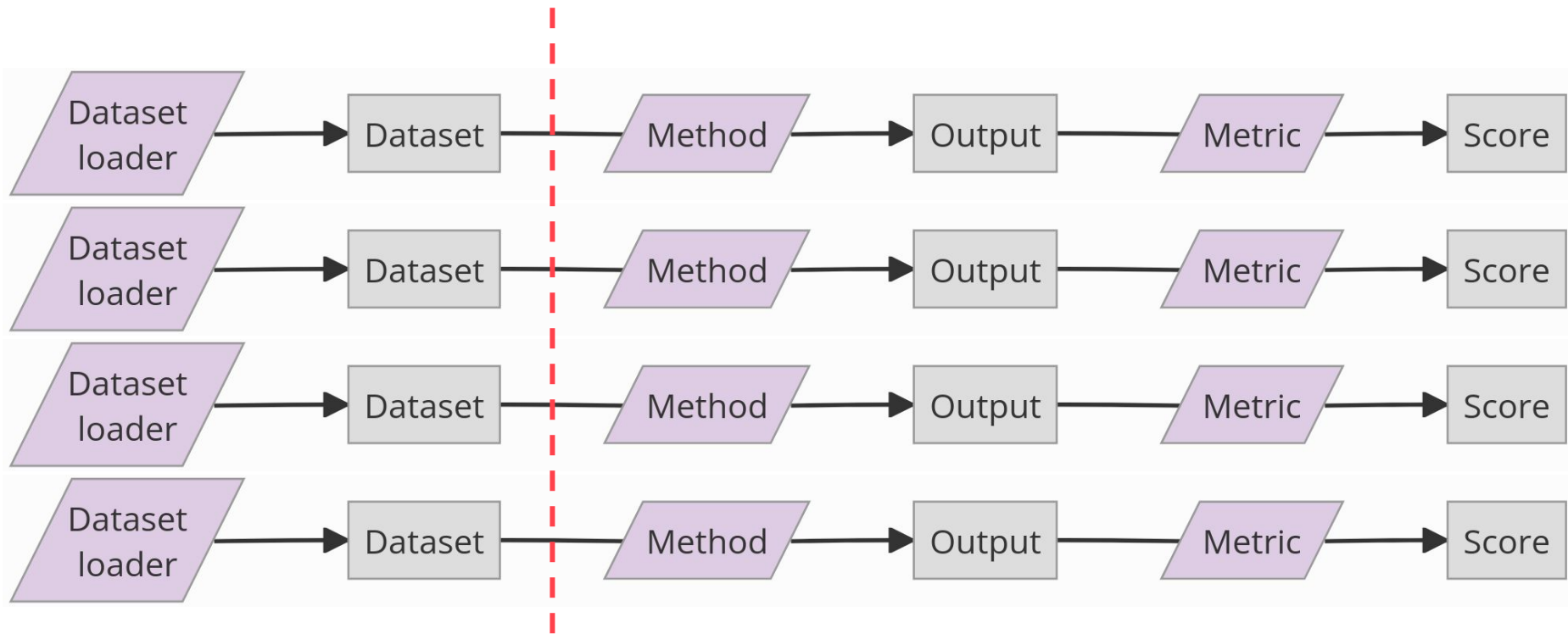
No reusability

Structure of pretty much any benchmark

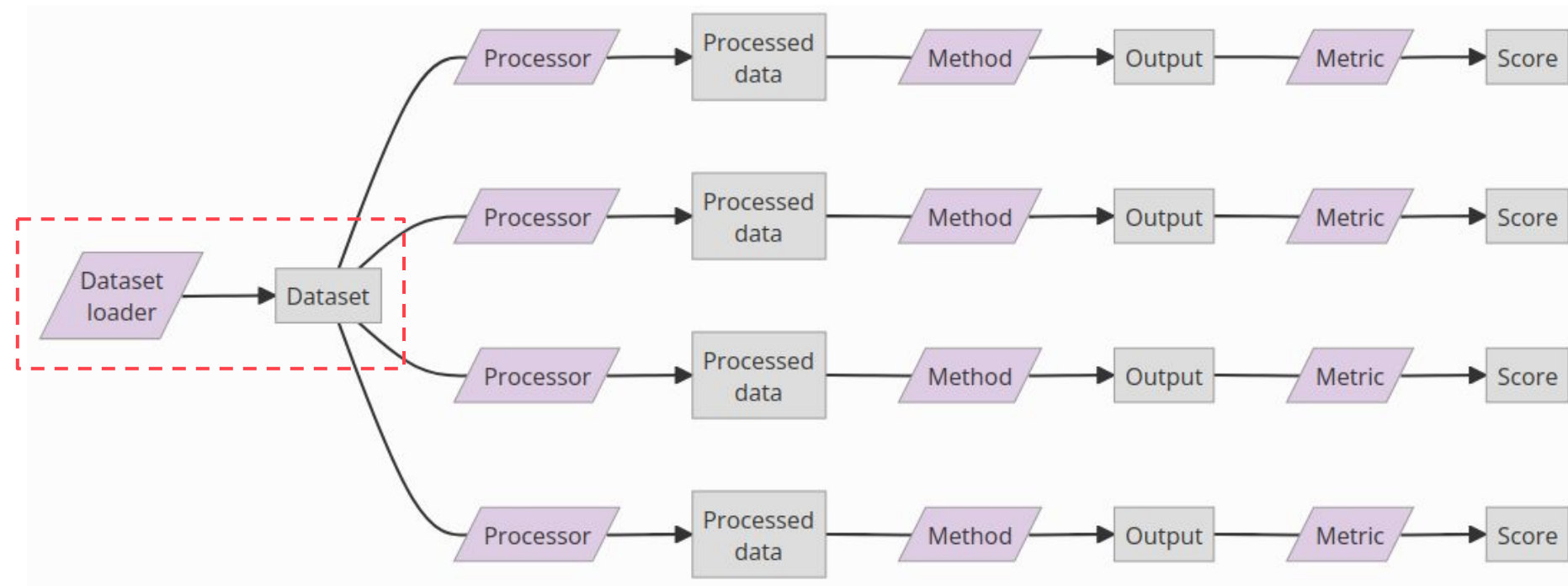


Legend: Grey rectangles are HDF5-backed [AnnData](#) (.h5ad) files. Purple rhomboids are [Viash](#) components.

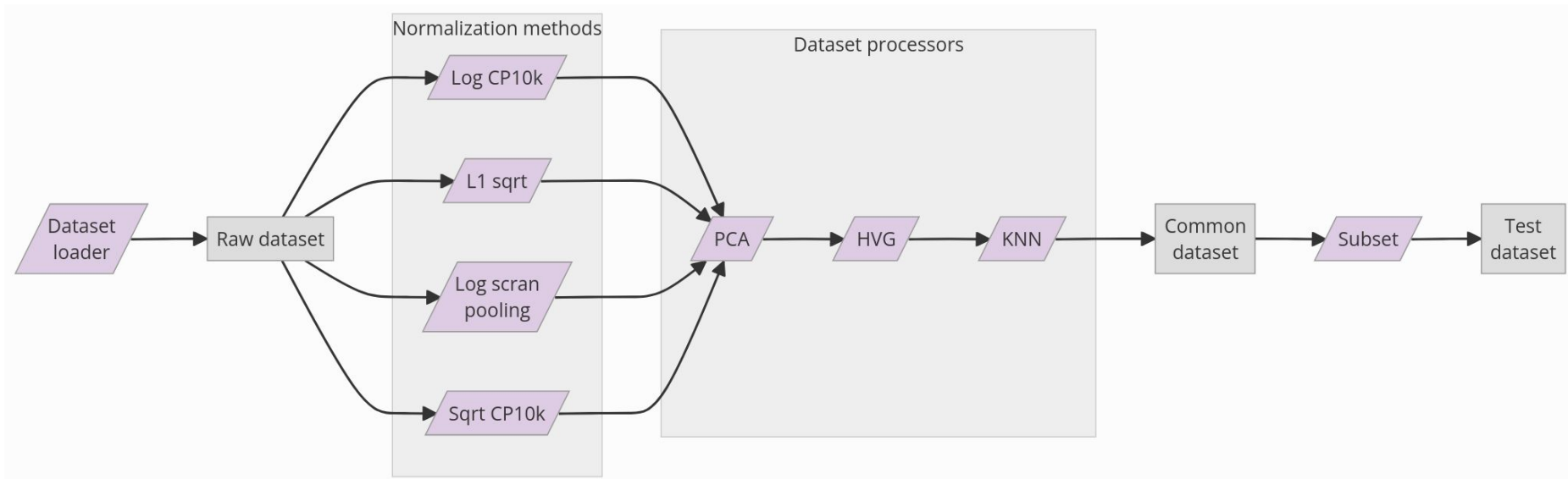
Structure of pretty much any benchmark



Reusable datasets



Common scRNA-seq dataset processing



Output: a **Common dataset** in the form of an .h5ad file

Current modalities: scRNA-seq, multi-omics (GEX+ADT, GEX+ATAC), and spatial Tx

All datasets¹ have a corresponding schema file²

This "Common dataset H5AD" schema is based on the [CELLxGENE schema](#) with some extra metadata.

[cellxgene_census/hlca](#) is an AnnData object with $n_obs \times n_vars = 584944 \times 28024$ with slots:

- `obs`: soma_joinid, dataset_id, assay, assay_ontology_term_id, cell_type, cell_type_ontology_term_id, development_stage, development_stage_ontology_term_id, disease, disease_ontology_term_id, donor_id, is_primary_data, self_reported_ethnicity, self_reported_ethnicity_ontology_term_id, sex, sex_ontology_term_id, suspension_type, tissue, tissue_ontology_term_id, tissue_general, tissue_general_ontology_term_id, batch, size_factors
- `var`: soma_joinid, feature_id, feature_name, hvg, hvg_score
- `obsp`: knn_connectivities, knn_distances
- `obsm`: X_pca
- `varm`: pca_loadings
- `layers`: counts, normalized
- `uns`: dataset_description, dataset_id, dataset_name, dataset_organism, dataset_reference, dataset_summary, dataset_url, knn, normalization_id, pca_variance

¹ h5ad, zarr, YAML, JSON, parquet, csv, tsv

² All input/output files for workflows. Data files in intermediary workflow steps might not have a schema associated with them.

File schema

Each input/output file has a specification of which slots are supposed to contain which information.

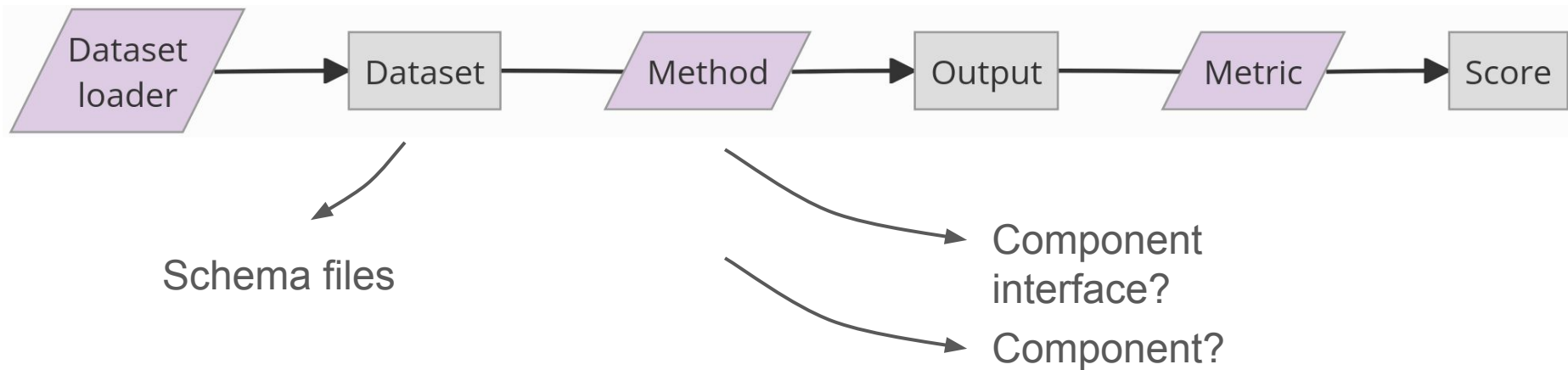
→ Used for documentation and unit testing

Example: [link](#)

```
type: file
example: "resources_test/task_label_projection/pancreas/dataset.h5ad"
label: "Training data"
summary: "A dataset containing the train cells and their labels."
info:
  format:
    type: h5ad
  layers:
    - type: integer
      name: counts
      description: Raw counts
      required: true
    - type: double
      name: normalized
      description: Normalized expression values
      required: true
  obs:
    - type: string
      name: cell_type
      description: Ground truth cell type labels
      required: true
  uns:
    - type: string
      name: dataset_id
      description: "A unique identifier for the dataset"
      required: true
    - type: string
      name: normalization_id
      description: "Which normalization was used"
      required: true
```

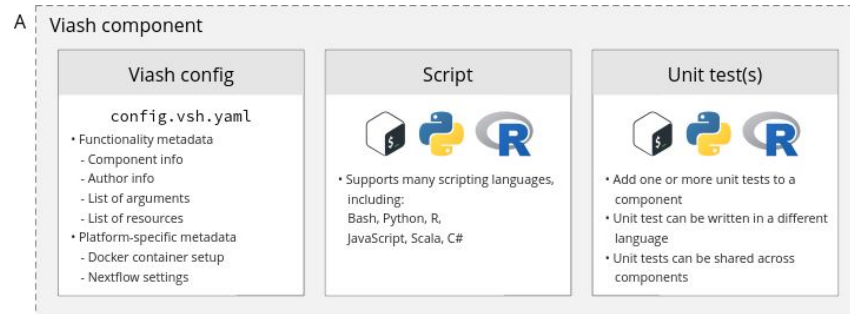
Dataset file schema: ✓

Component interfaces: ?



Legend: Grey rectangles are HDF5-backed [AnnData](#) (.h5ad) files. Purple rhomboids are [Viash](#) components.

Viash components (a bit like pandoc, but for scripts)



Example of a Viash component

```
# contents of 'config.vsh.yaml'
# Metadata
name: logistic_regression
namespace: methods
label: Logistic Regression
summary: Logistic Regression with [...]
description: |
  Logistic Regression estimates parameters [...]
references:
  doi: 123456/123456.1234.789456
links:
  repository: https://github.com/scikit-lea[...]
  documentation: https://scikit-learn.org/s [...]
# Interface
...
```

```
# contents of 'config.vsh.yaml' (cont)
arguments: # Interface
- name: "--input_train"
  type: file
  direction: input
  required: true
- name: "--input_test"
  type: file
  direction: input
  required: true
- name: "--output"
  type: file
  direction: output
  required: true
```

```
# script.py:
import anndata as ad
import sklearn.linear_model
import pandas as pd

train_adata = ad.read_h5ad(par["input_train"])
test_adata = ad.read_h5ad(par["input_test"])

# train model
classifier = \
    sklearn.linear_model.LogisticRegression()
classifier.fit(train_adata.obsm["X_pca"],
train_adata.obs["cell_type"].astype(str))

# create output
obs_label_pred = classifier.predict(
    test_adata.obsm["X_pca"]
)

# create output
output = ad.AnnData(
    obs = pd.DataFrame(
        { "pred": obs_label_pred },
        index = test_adata.index
    )
)
output.write_h5ad(par["output"])
```

Component API

Common settings such as input/output arguments, unit tests and test resources.

Example: [link](#)

```
namespace: methods
info:
  type: method
  type_info:
    label: Method
    summary: A label projection method.
    description: |
      A label projection method to predict the labels of a new "test"
      dataset based on an annotated "training" dataset.
arguments:
- name: "--input_train"
  __merge__: file_train.yaml
  direction: input
  required: true
- name: "--input_test"
  __merge__: file_test.yaml
  direction: input
  required: true
- name: "--output"
  __merge__: file_prediction.yaml
  direction: output
  required: true
test_resources:
- # paths to unit tests and test resource
```

Actual method config

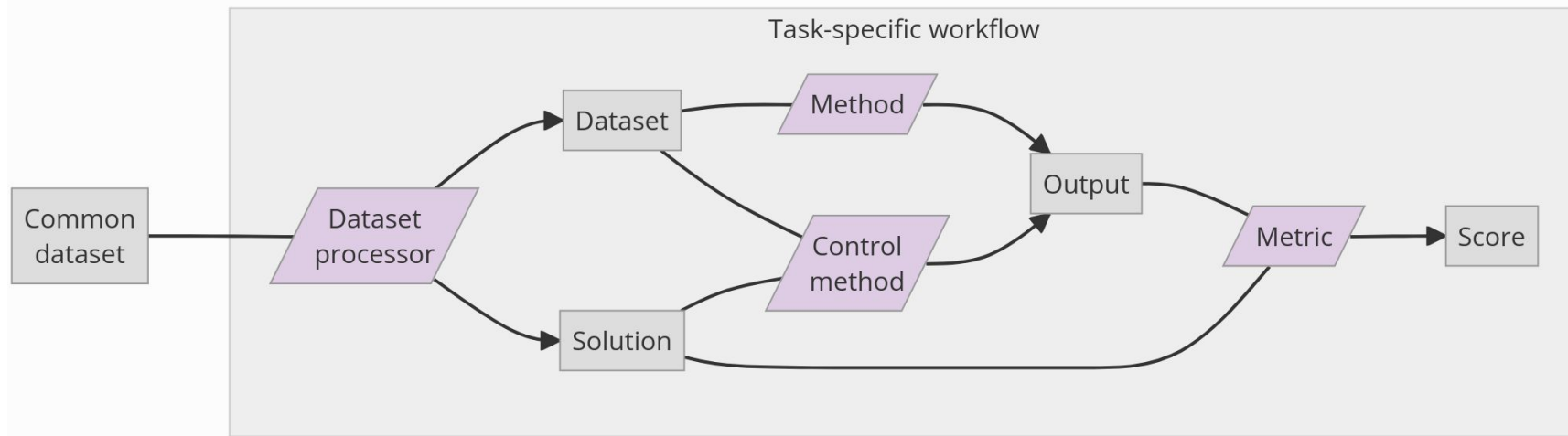
The amount of information a contributor needs to provide in a Viash config has been reduced as much as possible.

Example: [link](#)

Viash docs: viash.io

```
# Common settings such as input/output arguments, unit tests and test resources
__merge__: /src/api/comp_method.yaml
# Metadata
name: logistic_regression
label: Logistic Regression
summary: Logistic Regression with 100-dimensional PCA coordinates
description: |
  Logistic Regression estimates parameters of a logistic function for [...]
references:
  doi: 123456/123456.1234.789456
links:
  repository: https://github.com/scikit-learn/scikit-learn
  documentation: https://scikit-learn.org/stable/[...]
# Code
resources:
  - type: python_script
    path: script.py
# Reproducibility
engines:
  - type: docker
    image: openproblems/base_python:1.0.0
    setup:
      - type: python
        packages: scikit-learn
# Usability
runners:
  - type: executable
  - type: nextflow
  directives:
    label: [midtime, midmem, lowcpu]
```

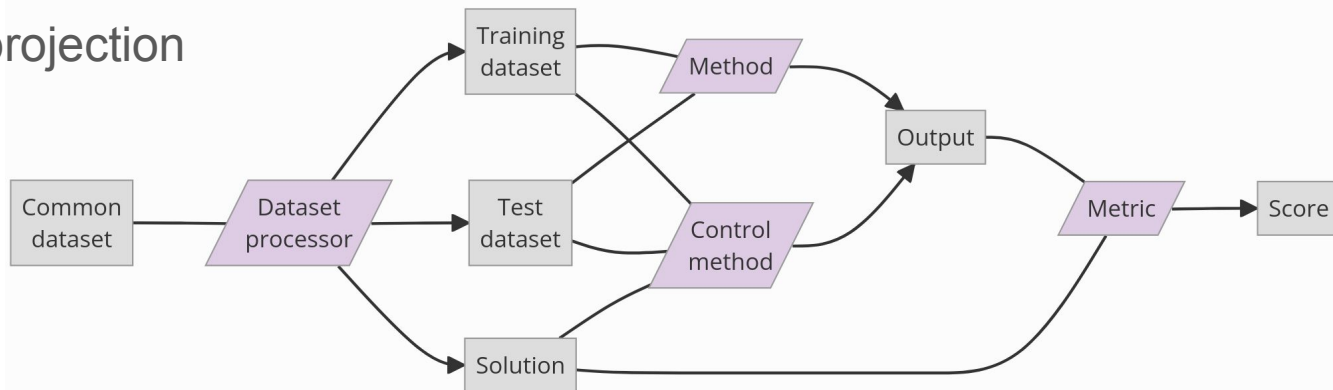
Structure of a typical OpenProblems task benchmark



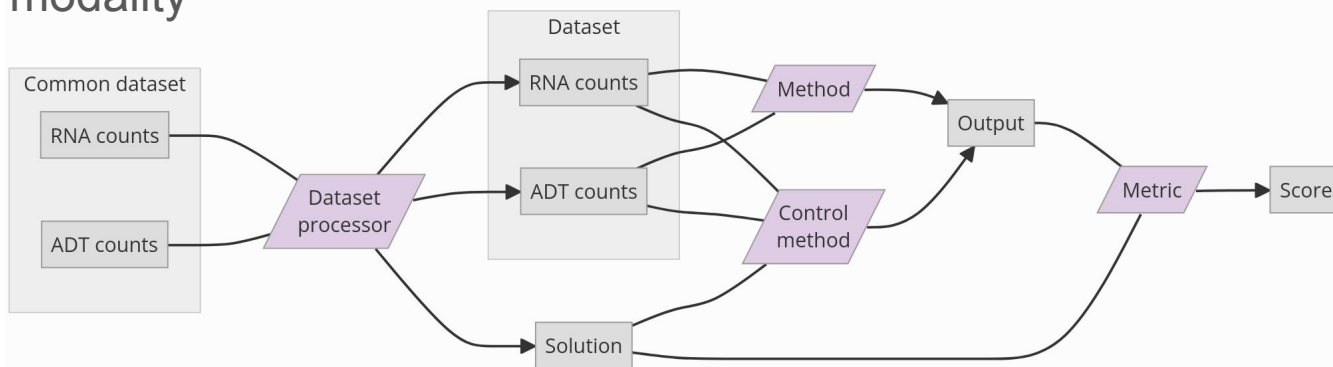
- **API**: The file schemas and component interfaces for each entity in the above graph
- **Data processor**: transforms a Common dataset to task-specific data objects
- **Method**: A computational algorithm to solve the task at hand
- **Metric**: A performance metric that compares a method output to the solution
- **Control methods**: Positive or negative control methods which are used for quality control of the task results and scaling of metric values

Topology of a task depends on experimental design

Label projection



Predict modality

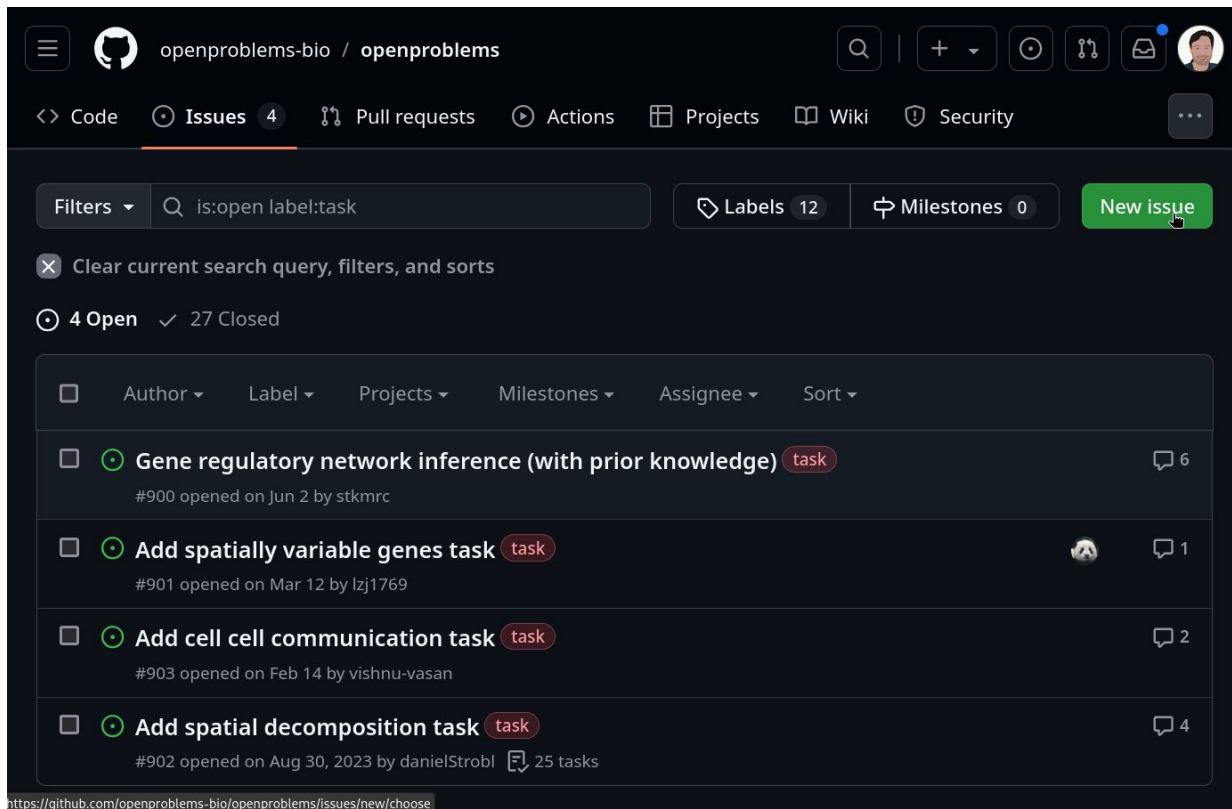


Safeguarding benchmark integrity

- Reuse **common datasets**: Can't introduce new bugs in preprocessing
- **Control methods**: Quality control of the benchmarking results
- **File schemas**: Automatically validate component outputs, detect bugs earlier
- Automated **unit testing**
- Automated **documentation**
- **CI CI CI**
- **Versioned releases** of components, workflows, and Docker images
- **Run anywhere**: locally, cloud (AWS, Google, Azure), HPC

Creating a new task

Step 1: Check whether a similar task already exists

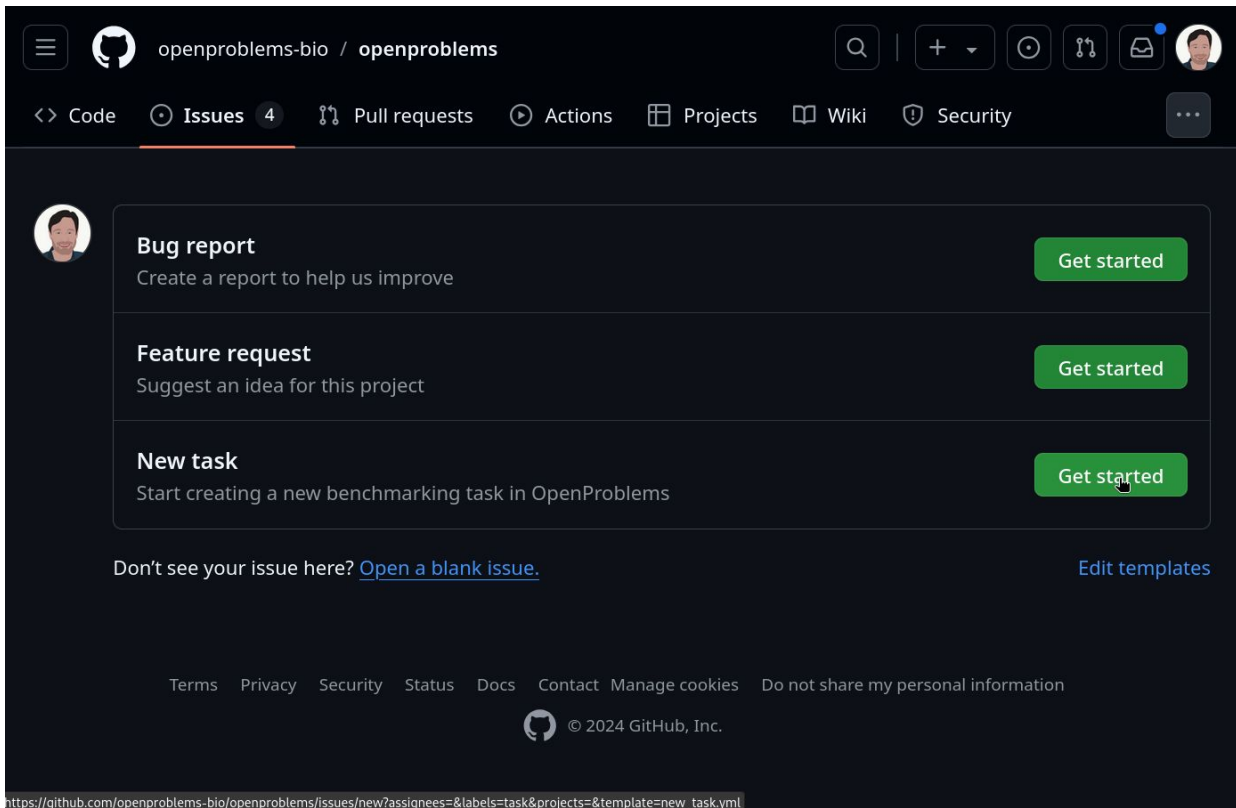


The screenshot shows the GitHub interface for the repository `openproblems-bio / openproblems`. The 'Issues' tab is selected, showing 4 open issues. The search bar contains the query `is:open label:task`. The issues listed are:

- ☐ **Gene regulatory network inference (with prior knowledge)** task
#900 opened on Jun 2 by stkmrc 6 comments
- ☐ **Add spatially variable genes task** task
#901 opened on Mar 12 by lzj1769 1 comment
- ☐ **Add cell cell communication task** task
#903 opened on Feb 14 by vishnu-vasan 2 comments
- ☐ **Add spatial decomposition task** task
#902 opened on Aug 30, 2023 by danielStrobl 25 tasks 4 comments

The URL at the bottom of the page is <https://github.com/openproblems-bio/openproblems/issues/new/choose>.

Step 2: Reach out to the OpenProblems team



The screenshot shows the GitHub interface for the 'openproblems-bio / openproblems' repository. The 'Issues' tab is selected, showing 4 issues. Below the repository name, there are three issue templates: 'Bug report', 'Feature request', and 'New task'. Each template has a 'Get started' button. At the bottom, there is a link to 'Open a blank issue' and a link to 'Edit templates'. The footer contains links for Terms, Privacy, Security, Status, Docs, Contact, Manage cookies, and Do not share my personal information, along with the GitHub logo and copyright notice.

openproblems-bio / openproblems

<> Code Issues 4 Pull requests Actions Projects Wiki Security

Bug report
Create a report to help us improve [Get started](#)

Feature request
Suggest an idea for this project [Get started](#)

New task
Start creating a new benchmarking task in OpenProblems [Get started](#)

Don't see your issue here? [Open a blank issue.](#) [Edit templates](#)

Terms Privacy Security Status Docs Contact Manage cookies Do not share my personal information

© 2024 GitHub, Inc.

https://github.com/openproblems-bio/openproblems/issues/new?assignees=&labels=task&projects=&template=new_task.yml

Step 2: Reach out to the OpenProblems team

Issue: New task

Start creating a new benchmarking task in OpenProblems. If this doesn't look right, [choose a different type](#).

**Add a title**

Thanks for choosing OpenProblems. Please check the [OpenProblems tasks](#) to see whether a similar task has already been created. If you haven't already, please review the documentation on [how to create a new task](#).

Task motivation

Explain the motivation behind your proposed task. Describe the biological or computational problem you aim to address and why it's important. Discuss the current state of research in this area and any gaps or challenges that your task could help address. This section should convince readers of the significance and relevance of your task.

(Basically the information from your feasibility study)

Task description

Provide a clear and concise description of your task, detailing the specific problem it

Assignees

No one—[assign yourself](#)

Labels

task

Projects

None yet

Milestone

No milestone

Development

Shows branches and pull requests linked to this issue.

Helpful resources

[Contributing](#)

[Code of conduct](#)

[GitHub Community Guidelines](#)

You're using an [issue form](#), a

Step 3: Create task repository

 New repository

 |     

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk ().*

Repository template

 openproblems-bio/task_template ▾

Start your repository with a template repository's contents.

☐ **Include all branches**
Copy all branches from openproblems-bio/task_template and not just the default branch.

Owner * **Repository name ***

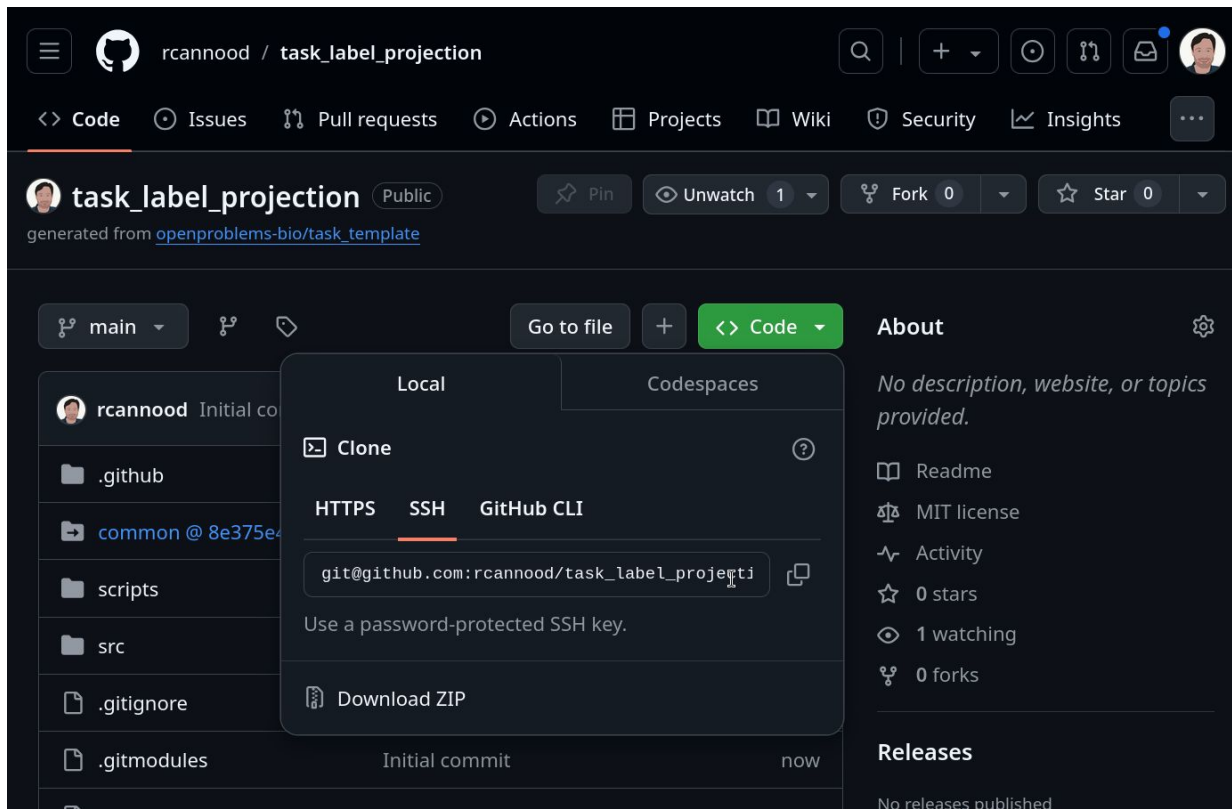
 rcannood ▾ / task_label_projection 

✔ task_label_projection is available.

Great repository names are short and memorable. Need inspiration? How about [turbo-computing-machine](#) ?

Description (optional)

Step 4: Clone the task repository



Step 4 & 5: Clone and test the task template

```
# Clone repository:
git clone --recursive git@github.com:rcannood/task_label_projection.git
cd task_label_projection

# Sync test resources
scripts/sync_resources.sh

# Run one component
viash run src/methods/logistic_regression/config.vsh.yaml -- --help

viash run src/methods/logistic_regression/config.vsh.yaml -- \
  --input_train resources_test/task_template/pancreas/train.h5ad \
  --input_test resources_test/task_template/pancreas/test.h5ad \
  --output output.h5ad

# Build all components:
viash ns build --parallel --setup cachedbuild

# Test one component:
viash test src/methods/logistic_regression/config.vsh.yaml

# Run the benchmarking workflow on the test datasets
scripts/run_benchmark/run_test_local.sh
```

Step 6: Look at repository structure

```
tree -L 2
├── CHANGELOG.md
├── common                # git submodule with common resources
├── CONTRIBUTING.md
├── LICENSE
├── main.nf
├── nextflow.config
├── README.md
├── resources_test        # test resources
│   ├── common            # common test resources you get after running sync_resources
│   └── task_template      # task-specific test resources you need to generate
├── scripts
│   ├── create_component  # generator script for creating a new method or metric
│   ├── create_readme.sh  # generate README from AI files
│   ├── create_resources  # scripts to generate test resources
│   ├── project           # build and test scripts
│   ├── run_benchmark     # run the benchmark
│   └── sync_resources.sh  # fetch test resources
├── src
│   ├── api               # the api yamls
│   ├── control_methods   # control method components
│   ├── data_processors    # dataset processor
│   ├── methods           # method components
│   ├── metrics           # metrics components
│   └── workflows         # viash nextflow workflow
└── _viash.yaml           # project config
```

Step 7: Update project config '_viash.yaml'

```
tree -L 2
├── CHANGELOG.md
├── common                # git submodule with common resources
├── CONTRIBUTING.md
├── LICENSE
├── main.nf
├── nextflow.config
├── README.md
├── resources_test        # test resources
│   ├── common            # common test resources you get after running sync_resources
│   └── task_template      # task-specific test resources you need to generate
├── scripts
│   ├── create_component  # generator script for creating a new method or metric
│   ├── create_readme.sh  # generate README from AI files
│   ├── create_resources  # scripts to generate test resources
│   ├── project           # build and test scripts
│   ├── run_benchmark     # run the benchmark
│   └── sync_resources.sh  # fetch test resources
├── src
│   ├── api               # the api yamls
│   ├── control_methods   # control method components
│   ├── data_processors   # dataset processor
│   ├── methods           # method components
│   ├── metrics           # metrics components
│   └── workflows         # viash nextflow workflow
└── _viash.yaml           # project config <-----
```

Step 7: Update project config '_viash.yaml'

```
name: task_label_projection

# documentation
label: Label Projection
summary: A one-sentence summary of the purpose and methodology of this benchmark.
description: |
  A multi-line description of the motivation behind this task and its experimental design.
  If you followed the previous steps, you can copy paste this information from 'Step 2'.
references:
  doi: [ 10.1234/123456.7890.123456 ]
keywords: [ scRNA-seq, label projection ]
authors:
  - name: John Doe
license: MIT

# technical information
# ...
```


Step 8: Create file schemas

```
tree -L 2 ✓
├── CHANGELOG.md
├── common                # git submodule with common resources
├── CONTRIBUTING.md
├── LICENSE
├── main.nf
├── nextflow.config
├── README.md
├── resources_test        # test resources
│   ├── common            # common test resources you get after running sync_resources
│   └── task_template      # task-specific test resources you need to generate
├── scripts
│   ├── create_component  # generator script for creating a new method or metric
│   ├── create_readme.sh  # generate README from AI files
│   ├── create_resources  # scripts to generate test resources
│   ├── project           # build and test scripts
│   ├── run_benchmark     # run the benchmark
│   └── sync_resources.sh  # fetch test resources
├── src
│   ├── api               # the api yamls <-----
│   ├── control_methods   # control method components
│   ├── data_processors   # dataset processor
│   ├── methods           # method components
│   ├── metrics           # metrics components
│   └── workflows         # viash nextflow workflow
└── _viash.yaml           # project config
```

Step 8: Create API

```
tree src/api
├─ comp_control_method.yaml
├─ comp_data_processor.yaml
├─ comp_method.yaml
├─ comp_metric.yaml
├─ file_common_dataset.yaml
├─ file_prediction.yaml
├─ file_score.yaml
├─ file_solution.yaml
├─ file_test_h5ad.yaml
└─ file_train_h5ad.yaml
```

```
1 directory, 10 files
```

Step 8: Create API - file schemas (src/api/file_train.yaml)

```
type: file
example: "resources_test/task_label_projection/pancreas/dataset.h5ad"
label: "Training data"
summary: "A dataset containing the train cells and their labels."
info:
  format:
    type: h5ad
  layers:
    - type: integer
      name: counts
      description: Raw counts
      required: true
    - type: double
      name: normalized
      description: Normalized expression values
      required: true
  obs:
    - type: string
      name: cell_type
      description: Ground truth cell type labels
      required: true
  uns:
    - type: string
      name: dataset_id
      description: "A unique identifier for the dataset"
      required: true
    - type: string
      name: normalization_id
      description: "Which normalization was used"
      required: true
```

Step 8: Create API - component interface (src/api/comp_method.yaml)

```
namespace: methods
info:
  type: method
  type_info:
    label: Method
    summary: A label projection method.
    description: |
      A label projection method to predict the labels of a new "test"
      dataset based on an annotated "training" dataset.
arguments:
  - name: "--input_train"
    __merge__: file_train.yaml
    direction: input
    required: true
  - name: "--input_test"
    __merge__: file_test.yaml
    direction: input
    required: true
  - name: "--output"
    __merge__: file_prediction.yaml
    direction: output
    required: true
test_resources:
  - # paths to unit tests and test resources
```

Step 9: Render README

scripts/create_readme.sh

READMEMIT license

Label projection

Automated cell type annotation from rich, labeled reference data

Repository: [openproblems-bio/task_label_projection](#)

Description

A major challenge for integrating single cell datasets is creating matching cell type annotations for each cell. One of the most common strategies for annotating cell types is referred to as [“cluster-then-annotate”](#) whereby cells are aggregated into clusters based on feature similarity and then manually characterized based on differential gene expression or previously identified marker genes. Recently, methods have emerged to build on this strategy and annotate cells using [known marker genes](#). However, these strategies pose a difficulty for integrating atlas-scale datasets as the particular annotations may not match.

To ensure that the cell type labels in newly generated datasets match existing reference datasets, some methods align cells to a previously annotated [reference dataset](#) and then *project* labels from the reference to the new dataset.

Here, we compare methods for annotation based on a reference dataset. The datasets consist of two or more samples of single cell profiles that have been manually annotated with matching labels. These datasets are then split into training and test batches, and the task of each method is to train a cell type classifier on the training set and project those labels onto the test set.

Authors & contributors

name	roles
Nikolay Markov	author, maintainer
Scott Gigante	author
Robrecht Cannoodt	author

API

```
graph LR; CD[Common Dataset] --> DP[/Data processor/]; DP --> S[Solution]; DP --> TD[Test data]; DP --> TRD[Training data]; S --> CM[/Control method/]; TD --> CM; TRD --> M[/Method/]; CM --> P[Prediction]; M --> P; P --> ME[/Metric/]; ME --> SC[Score];
```

File format: Common Dataset

A subset of the common dataset.

Example file: `resources_test/common/pancreas/dataset.h5ad`

Format:

```
AnnData object
obs: 'cell_type', 'batch'
var: 'hvg', 'hvg_score'
obsm: 'X_pca'
layers: 'counts', 'normalized'
uns: 'dataset_id', 'dataset_name', 'dataset_url', 'dataset_reference', 'dataset_summary', 'da
```

Data structure:

Slot	Type	Description
<code>obs["cell_type"]</code>	string	Cell type information.
<code>obs["batch"]</code>	string	Batch information.
<code>var["hvg"]</code>	boolean	Whether or not the feature is considered to be a 'highly variable gene'.

Step 10: Create components

```
tree -L 2 ✓
├── CHANGELOG.md
├── common                # git submodule with common resources
├── CONTRIBUTING.md
├── LICENSE
├── main.nf
├── nextflow.config
├── README.md
├── resources_test        # test resources
│   ├── common            # common test resources you get after running sync_resources
│   └── task_template      # task-specific test resources you need to generate
├── scripts
│   ├── create_component  # generator script for creating a new method or metric
│   ├── create_readme.sh  # generate README from AI files
│   ├── create_resources  # scripts to generate test resources
│   ├── project           # build and test scripts
│   ├── run_benchmark     # run the benchmark
│   └── sync_resources.sh  # fetch test resources
├── src
│   ├── api               # the api yamls
│   ├── control_methods   # control method components <-----
│   ├── data_processors   # dataset processor <-----
│   ├── methods           # method components <-----
│   ├── metrics           # metrics components <-----
│   └── workflows         # viash nextflow workflow
└── _viash.yaml           # project config
```

Step 10: Create components

tree -L 2



```
├── CHANGELOG.md
├── common
├── CONTRIBUTING.md
├── LICENSE
├── main.nf
├── nextflow.config
├── README.md
├── resources_test
│   ├── common
│   └── task_template
├── scripts
│   ├── create_component
│   ├── create_readme.sh
│   ├── create_resources
│   ├── project
│   ├── run_benchmark
│   └── sync_resources.sh
├── src
│   ├── api
│   ├── control_methods
│   ├── data_processors
│   ├── methods
│   ├── metrics
│   └── workflows
└── _viash.yaml
```

```
import anndata as ad
```

```
# Fetch dataset from somewhere.
```

```
adata = ad.read_h5ad(\
    "resources_test/common/cxg_mouse_pancreas_atlas/dataset.h5ad")
```

```
# Process dataset: Divide the dataset into train and test sets.
```

```
test_batch = adata.obs["batch"].to_list()[0]
train_adata = adata[adata.obs["batch"] != test_batch].copy()
test_adata = adata[adata.obs["batch"] == test_batch].copy()
```

```
# Method: Apply logistic regression
```

```
import sklearn.linear_model
classifier = sklearn.linear_model.LogisticRegression()
classifier.fit(train_adata.obsm["X_pca"],
train_adata.obs["cell_type"].astype(str))
obs_label_pred = classifier.predict(test_adata.obsm["X_pca"])
```

```
# Metric: Compute accuracy
```

```
import numpy as np
accuracy = np.mean(test_adata.obs["cell_type"] == obs_label_pred)
accuracy # 0.9478672985781991
```

Step 10: Create components

```
# Component interface
__merge__: /src/api/comp_method.yaml
# Metadata
name: logistic_regression
label: Logistic Regression
summary: Logistic Regression with 100-dimensional PCA coordinates
description: |
    Logistic Regression estimates parameters of a logistic function for [...]
references:
    doi: 123456/123456.1234.789456
links:
    repository: https://github.com/scikit-learn/scikit-learn
    documentation: https://scikit-learn.org/stable/[...]
# Code
resources:
    - type: python_script
      path: script.py
# Reproducibility
engines:
    - type: docker
      image: openproblems/base_python:1.0.0
      setup:
        - type: python
          packages: scikit-learn
# Usability
runners:
    - type: executable
    - type: nextflow
directives:
    label: [midtime, midmem, lowcpu]
```

```
import anndata as ad
import sklearn.linear_model
import pandas as pd

print('Reading input files', flush=True)
input_train = ad.read_h5ad(par['input_train'])
input_test = ad.read_h5ad(par['input_test'])

print('Train model', flush=True)
classifier = sklearn.linear_model.LogisticRegression()
classifier.fit(input_train.obsm["X_pca"], input_train.obs["label"].astype(str))

print('Generate predictions', flush=True)
obs_label_pred = classifier.predict(input_test.obsm["X_pca"])

print("Create output data structure", flush=True)
output = ad.AnnData(
    obs=pd.DataFrame(
        { 'label_pred': obs_label_pred },
        index=input_test.obs.index,
    ),
    uns={
        'dataset_id': input_train.uns['dataset_id'],
        'normalization_id': input_train.uns['normalization_id'],
        'method_id': meta['name']
    },
)

print("Write output AnnData to file", flush=True)
output.write_h5ad(par['output'], compression='gzip')
```


Step 11: Build executables and workflows

```
tree -L 2 ✓
├── CHANGELOG.md
├── common                # git submodule with common resources
├── CONTRIBUTING.md
├── LICENSE
├── main.nf
├── nextflow.config
├── README.md
├── resources_test        # test resources
│   ├── common            # common test resources you get after running sync_resources
│   └── task_template      # task-specific test resources you need to generate
├── scripts
│   ├── create_component  # generator script for creating a new method or metric
│   ├── create_readme.sh  # generate README from AI files
│   ├── create_resources  # scripts to generate test resources
│   ├── project           # build and test scripts
│   ├── run_benchmark     # run the benchmark
│   └── sync_resources.sh  # fetch test resources
├── src
│   ├── api               # the api yamls
│   ├── control_methods   # control method components
│   ├── data_processors   # dataset processor
│   ├── methods           # method components
│   ├── metrics           # metrics components
│   └── workflows         # viash nextflow workflow
├── target
│   ├── executable        # standalone executables <-----
│   └── nextflow          # nextflow modules & modules <-----
└── _viash.yaml           # project config
```

Step 11: Build executables and workflows

```
scripts/project/build_all_components.sh
# Exporting true_labels (control_methods)      =nextflow=> target/nextflow/control_methods/true_labels
# Exporting process_datasets (workflows)       =nextflow=> target/nextflow/workflows/process_datasets
# Exporting run_benchmark (workflows)          =nextflow=> target/nextflow/workflows/run_benchmark
# Exporting logistic_regression (methods)      =nextflow=> target/nextflow/methods/logistic_regression
# Exporting accuracy (metrics)                 =nextflow=> target/nextflow/metrics/accuracy
# Exporting process_dataset (data_processors)  =nextflow=> target/nextflow/data_processors/process_dataset
# All 10 configs built successfully
```

```
tree target
# target
# ├── executable
# |   ├── control_methods/true_labels/true_labels
# |   ├── data_processors/process_dataset
# |   |   ├── process_dataset
# |   |   └── subset_h5ad_by_format.py
# |   ├── methods/logistic_regression/logistic_regression
# |   └── metrics/accuracy/accuracy
# └── nextflow
#     ├── control_methods/true_labels
#     |   ├── main.nf
#     |   └── nextflow.config
#     ├── data_processors/process_dataset/[...]
#     ├── methods/logistic_regression/[...]
#     ├── metrics/accuracy/[...]
#     └── workflows
#         ├── process_datasets/[...]
#         └── run_benchmark/[...]
```

Step 12: Adapt workflow and run it

```
scripts/run_benchmark/run_test_local.sh
```

```
# Launching `target/nextflow/workflows/run_benchmark/main.nf` [lonely_hilbert] DSL2 - revision: 5e77baf437
#
# executor > local (1)
# [56/a5bc85] aut...ract_uns_metadata:processWf:extract_uns_metadata_process (pancreas) [100%] 1 of 1 ✓
# [a9/503cfc] aut...Wf:true_labels:processWf:true_labels_process (pancreas.true_labels) [100%] 1 of 1 ✓
# [29/8a3b51] aut...rocessWf:logistic_regression_process (pancreas.logistic_regression) [100%] 1 of 1 ✓
# [6f/4adf41] aut...accuracy:processWf:accuracy_process (pancreas.true_labels.accuracy) [100%] 2 of 2 ✓
# [b5/1590ec] aut...es:processWf:extract_scores_process (pancreas.true_labels.accuracy) [100%] 2 of 2 ✓
# [0a/8be232] auto:run_benchmark:publishStatesSimpleWf:publishStatesProc (output) [100%] 1 of 1 ✓
```

```
tree resources/results/testrun_2024-09-11_13-41-59
```

```
# resources/results/testrun_2024-09-11_13-41-59
# |— dataset_uns.yaml
# |— method_configs.yaml
# |— metric_configs.yaml
# |— score_uns.yaml
# |— state.yaml
# |— task_info.yaml
#
# 1 directory, 6 files
```

After a local run on
test data **succeeds**,
we run in on our
AWS batch

< View Workflow Run

 **focused_davinci** [workflows/run_benchmark](#) 

full task_label_projection 

Command lineParametersConfigurationDatasetsExecution logReports

```
nextflow run 'https://github.com/openproblems-bio/task_label_projection.git'  
-name focused_davinci  
-params-file 'https://api.cloud.seqera.io/ephemeral/PikqVN186Nx9JYfgQBERWw.yaml'  
-with-tower  
-r build/main  
-entry auto  
-main-script target/nextflow/workflows/run_benchmark/main.nf  
-latest
```

General

id 53YEqa5LOD2IQg

 focused_davinci

 2024-09-07 23:27:48

 e1db473a7e365acef1014c3b7764ffb78b7a06ce (build/main)

 7fd66b93-1e06-42dd-b1b7-ffbeae462b5b

 robrecht

 s3://openproblems-nextflow/work

 -

Status

0	0	0
pending	submitted	running
0	649	20
cached	succeeded	failed

After **results** are generated,
a **results page** on the website

 **Open Problems** in
Single-Cell Analysis

AboutDatasetsBenchmarksEventsTeamDocumentation



Label Projection

Automated cell type annotation from rich, labeled reference data

8 datasets · 16 methods · 2 control methods · 3 metrics

Task infoMethod infoMetric infoDataset infoResults

A major challenge for integrating single cell datasets is creating matching cell type annotations for each cell. One of the most common strategies for annotating cell types is referred to as “**cluster-then-annotate**” whereby cells are aggregated into clusters based on feature similarity and then manually characterized based on differential gene expression or previously identified marker genes. Recently, methods have emerged to build on this strategy and annotate cells using **known marker genes**. However, these strategies pose a difficulty for integrating atlas-scale datasets as the particular annotations may not match.

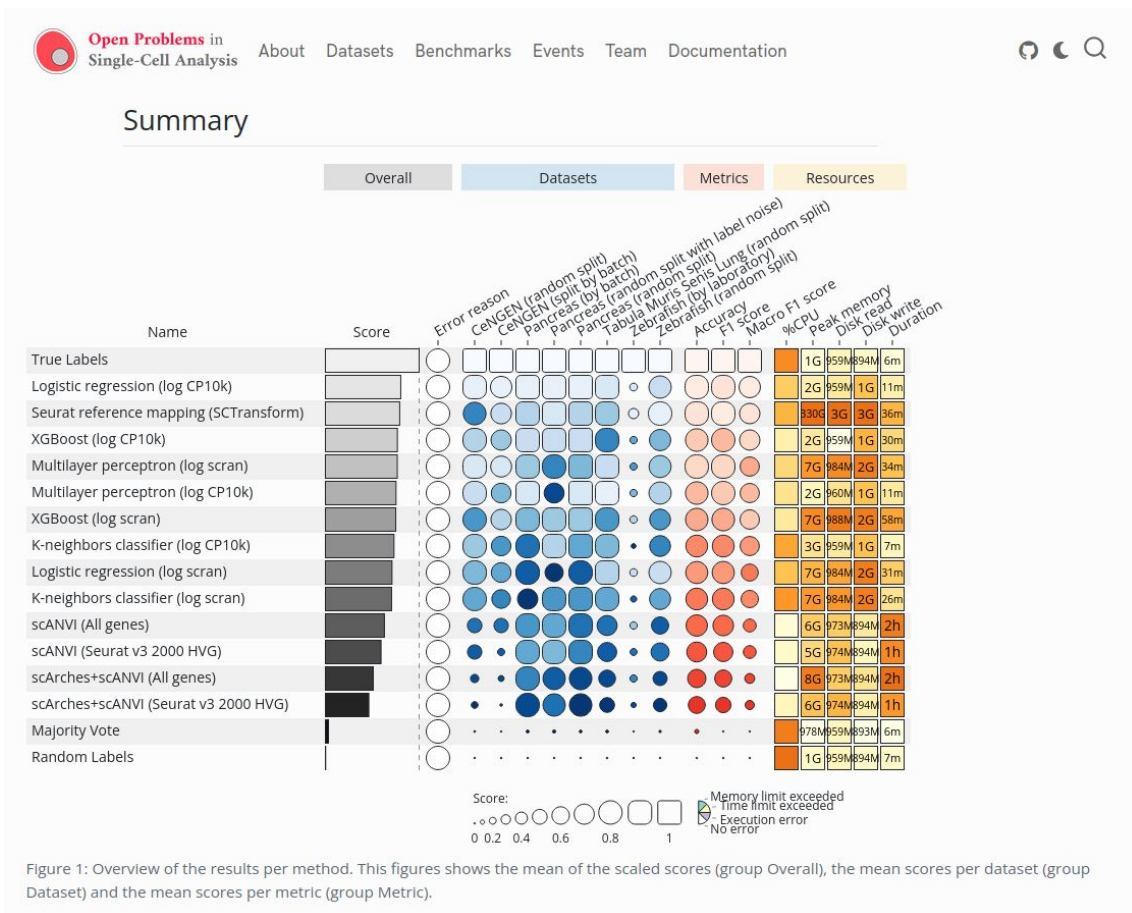
To ensure that the cell type labels in newly generated datasets match existing reference datasets, some methods align cells to a previously annotated **reference dataset** and then *project* labels from the reference to the new dataset.

Here, we compare methods for annotation based on a reference dataset. The datasets consist of two or more samples of single cell profiles that have been manually annotated with matching labels. These datasets are then split into training and test batches, and the task of each method is to train a cell type classifier on the training set and project those labels onto the test set.

Summary

OverallDatasetMethodResource

After **results** are generated,
a **results page** on the website

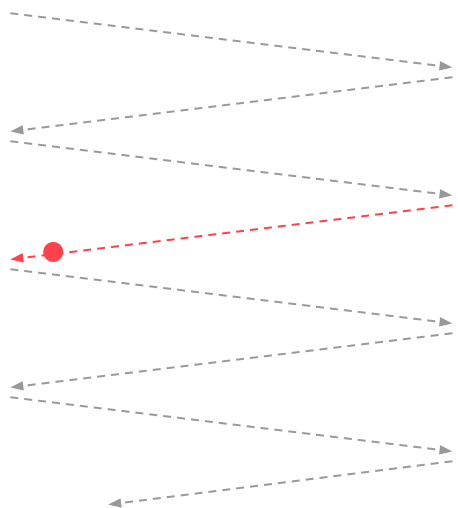


Research

- Feasibility study
- Literature study
- Preprint
- Write manuscript
- Publication

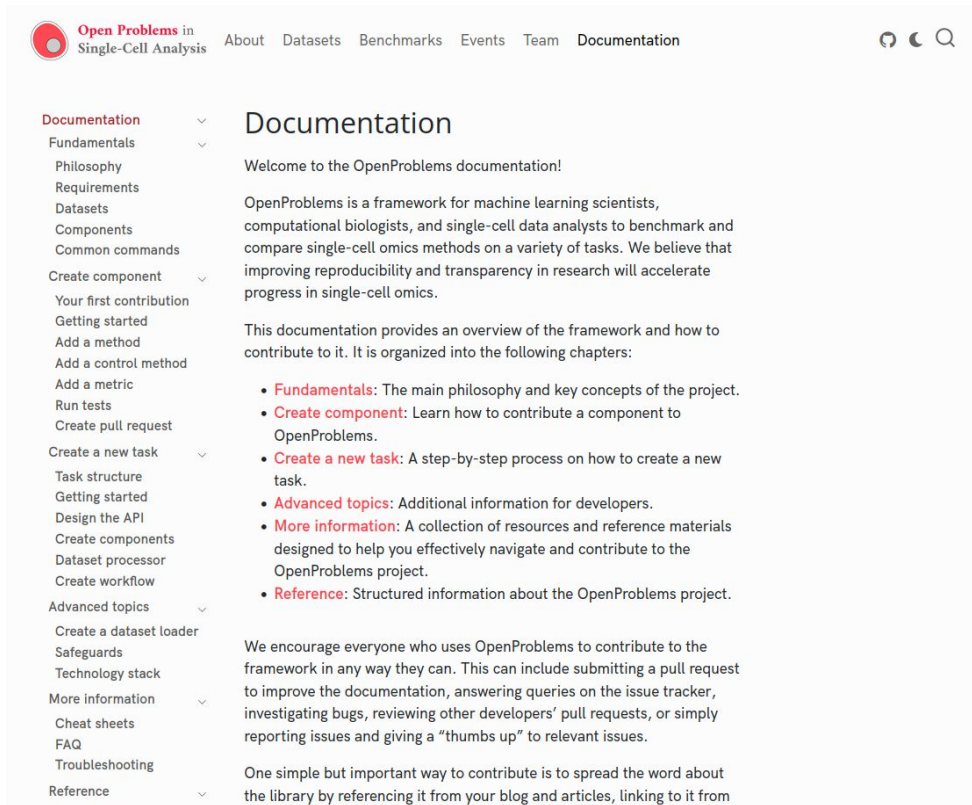
Coding

- Proof of concept
- Minimum viable product
- Production benchmark
- Finishing touches



This ended up not being an interactive workshop...

- Documentation at openproblems.bio/documentation
- Recordings of the full workshop planned
- Community involvement:
 - Discord
 - Weekly Friday working meetings
 - openproblems.bio/events/weekly_coworking



The screenshot shows the OpenProblems documentation website. The header includes the logo and navigation links: About, Datasets, Benchmarks, Events, Team, and Documentation. The left sidebar contains a table of contents with expandable sections: Documentation, Fundamentals, Create component, Create a new task, Advanced topics, More information, and Reference. The main content area is titled 'Documentation' and contains a welcome message, an overview of the project, and a list of chapters: Fundamentals, Create component, Create a new task, Advanced topics, More information, and Reference. The footer encourages users to contribute to the framework.

Open Problems in Single-Cell Analysis

About Datasets Benchmarks Events Team Documentation

Documentation

Welcome to the OpenProblems documentation!

OpenProblems is a framework for machine learning scientists, computational biologists, and single-cell data analysts to benchmark and compare single-cell omics methods on a variety of tasks. We believe that improving reproducibility and transparency in research will accelerate progress in single-cell omics.

This documentation provides an overview of the framework and how to contribute to it. It is organized into the following chapters:

- **Fundamentals:** The main philosophy and key concepts of the project.
- **Create component:** Learn how to contribute a component to OpenProblems.
- **Create a new task:** A step-by-step process on how to create a new task.
- **Advanced topics:** Additional information for developers.
- **More information:** A collection of resources and reference materials designed to help you effectively navigate and contribute to the OpenProblems project.
- **Reference:** Structured information about the OpenProblems project.

We encourage everyone who uses OpenProblems to contribute to the framework in any way they can. This can include submitting a pull request to improve the documentation, answering queries on the issue tracker, investigating bugs, reviewing other developers' pull requests, or simply reporting issues and giving a "thumbs up" to relevant issues.

One simple but important way to contribute is to spread the word about the library by referencing it from your blog and articles, linking to it from

Roadmap

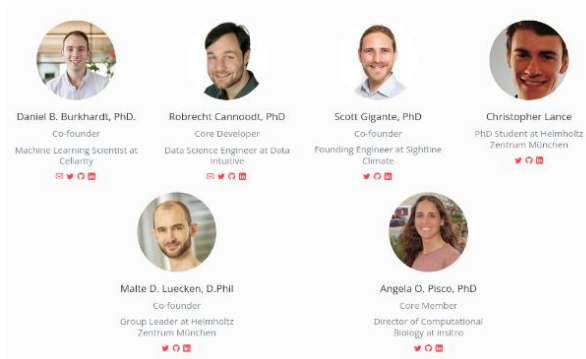
- Fellowship program
- Jamborees coming soon
- Changes to organisation structure:
 - Restructure common datasets per modality
 - Migrating benchmarks
from [openproblems-bio/openproblems](https://openproblems-bio.github.io/openproblems)
to [openproblems-bio/task_](https://openproblems-bio.github.io/task_)*
- Overhaul website:
 - New design
 - Backend DB
 - Versioned task results



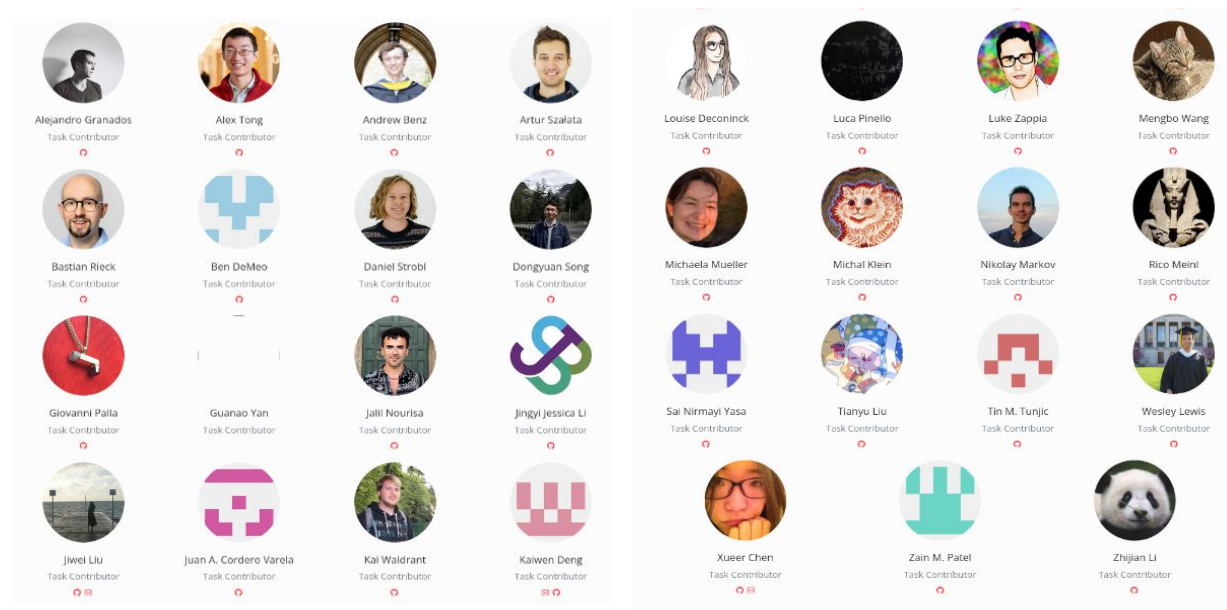
Join our Discord

Acknowledgements

Core team



Task contributors¹



Scientific advisors



Sponsors



¹: Reach out if your name is missing!